



Designer Reference Guide

Yoakum District Guidance

November 2024



Texas Department of Transportation

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General Information

1.1 Preface

These guidelines are maintained by the Yoakum District Transportation Design and Advanced Planning Office and are designed to provide a comprehensive, standardized approach to the planning and design of transportation projects within the Yoakum District. The objective of this document is to serve as a reference tool for designers, engineers, and project managers, ensuring consistent and efficient development of projects from initiation through final design and construction.

The guidelines focus on providing a unified framework for project development, with an emphasis on quality control, compliance with state and federal standards, and effective communication among all project stakeholders. Adherence to these standards will help ensure that all projects within the Yoakum District meet the required design specifications, are completed on schedule, and are executed within budget.

This document is intended for use by designers working within the Yoakum District as well as contractors, consultants, and other external entities engaged in the planning, design, and construction of district projects.

1.2 Project Development Manual/Checklist

The Project Development Manual/Checklist is a vital tool for managing the design and development of projects in the Yoakum District. The completion of this checklist is required for all projects, regardless of scale or complexity. This checklist serves as a step-by-step guide, ensuring that all necessary tasks and approvals are completed at each phase of the project.

- Purpose: The checklist is intended to help designers and project managers track the progress of a project from initial planning through final design and construction. It ensures that each phase of the project is thoroughly reviewed, documented, and aligned with district standards.
- Location and Access: The checklist is available as an Excel file and can be accessed through the Yoakum District's official standards and guidance page on the TxDOT website.
- Scoping Meeting: The Scoping Meeting should take place early in the project development process to define the project's scope, deliverables, and objectives. During this meeting, project managers,

engineers, planners, and any necessary stakeholders discuss the preliminary design requirements, project limits, resource needs, and timeline. The goal is to ensure that everyone has a shared understanding of the project's scope, identify any potential constraints, and address concerns that may arise during design or construction.

- Required for All Projects: Completion of the checklist is mandatory for all projects within the Yoakum District, regardless of project type or scale. The checklist should be updated regularly throughout the project development lifecycle to track progress and confirm that all required actions have been taken.
- Kick-off Meeting: The checklist should be reviewed and discussed during the Design Kick-off Meeting to ensure alignment between all project stakeholders, including engineers, planners, environmental specialists, and any external partners or contractors. The kick-off meeting marks the formal start of the project design phase and is a key opportunity to clarify the scope, objectives, and timelines.
- Safety Risk Mitigation Workshop: A Safety Meeting should be scheduled as part of the initial design discussions, ensuring that safety considerations are integrated into the project from the outset. This meeting focuses on identifying potential safety hazards, ensuring compliance with safety standards, and incorporating safety protocols into the design and construction phases. All stakeholders should participate, including safety officers, project managers, and design engineers, to discuss site-specific safety concerns and establish a comprehensive safety plan for the project.
- Pre-Design Review: It is essential that the Project Development Manual/Checklist be reviewed and completed before the Design Kick-off Meeting. This ensures that the project is properly scoped, all required steps are outlined, and potential issues are identified before work begins.

1.3 PS&E Development Stages

The Plans, Specifications, and Estimates (PS&E) development process is a structured series of stages that guide the design and preparation of a project for construction. Each stage builds upon the last, progressively refining the project's design, scope, and construction feasibility. The following stages outline the general progression of PS&E development, from initial scoping through to project letting.

0% - Scoping Meeting: At this stage, the team agrees on the overall project objectives and outlines the fundamental scope of work. Additionally, DSR is the predesign agenda for the most of our projects as it is used for larger, more complex projects. This stage ensures that

all key stakeholders have a clear understanding of the project's goals and requirements. The focus here is on defining the project and understanding its core elements. The outcome of this meeting is a shared vision and commitment to the project's direction.

Key Activities:

- Review project background and objectives
- Identify major design elements and constraints
- Establish deliveries
- Establish project timeline and budget overview

30% - Preliminary Design: At the 30% design stage, the project is beginning to take shape. The design team focuses on creating the basic framework of the project, which includes major alignments, key dimensions, and foundational layout elements. This stage provides a clear vision of what the final project will look like in terms of its general design and layout. While not all details are fleshed out, this stage sets the stage for more detailed design work.

Key Activities:

- Develop initial design horizontal and vertical alignments
- Preliminary assessment of environmental impacts and constraints
- Determine if project will fit within the existing ROW
- Assessing any possibilities of flooding issues
- Refine project cost estimates and construction phasing
- Review design criteria and standards
- Review impacts to utilities
- Replace or widen exist structures

- ✓ It is strongly recommended, if applicable, to submit the PBLR and/or steel structure details to the BRG Division when the design reaches between 30% and 60% completion.

60% - Intermediate Design: By the 60% design stage, the project's design has significantly matured. At this point, the team focuses on how the project will be built. This includes refining structural, environmental, and construction-related details, as well as identifying potential challenges in execution. The design is now closer to a full implementation plan,

with most major elements in place. Detailed construction documents and materials specifications start to take shape.

Key Activities:

- Finalize design of structural elements (e.g., bridges, culverts, drainage systems)
 - Confirm construction methodologies and traffic control plans
 - Perform detailed cost analysis and refine construction schedules
 - Ensure coordination with utilities, environmental agencies, and other stakeholders
- ✓ If applicable, a TCP Workshop is recommended for larger, more complex projects when the submittal reaches between 60% and 95%.

95% - Final Design and Project Management: At the 95% stage, the project is nearly ready for letting. This stage focuses on completing the final design and associated project management activities, ensuring that all the details are fully covered, and the project is prepared for execution. This includes ensuring compliance with all relevant regulatory and safety standards, finalizing paperwork, and obtaining approvals for the project.

Key Activities:

- Complete final design documentation, including specifications and drawings
- Perform final quality assurance and compliance checks
- Prepare bid documents, contracts, and permits
- Final review and approval of cost estimates and schedule

100% - Ready for Letting: At 100%, the project is fully ready for letting. All design work is complete, and the project is ready for construction bids. This stage can be compared to preparing for a trip—where all preparations are finalized, and the project is ready to go. At this point, all documents are in place for the construction phase, and the project is set to be advertised to contractors for bidding.

Key Activities:

- Finalize and submit PS&E package for bidding
- Ensure all documents are in compliance with state and federal regulations
- Confirm funding and scheduling for construction

- Publicize the project for contractor bidding

1.4 How to Scope a Project

Before the Scoping Meeting, it is essential to conduct a pre-scope assessment with the Maintenance Section Foreman and Area Engineer (AE). This visit serves to identify existing roadway issues, such as accident-prone areas, drainage problems, and pavement deterioration, and to discuss potential solutions. Key issues, such as railroad and Right of Way impacts, pavement rehabilitation options or drainage improvements, should be identified early.

Document the site conditions thoroughly by taking photographs and video footage during the visit. This documentation will be critical for refining the project scope and preparing accurate estimates. It is also important to note specific locations of concern, such as intersections with high accident rates or areas with significant drainage issues.

Following the site visit, develop a preliminary scope of work based on the findings. Summarize the issues and proposed solutions and use available as-built plans and right-of-way (ROW) maps to assist with the estimate. This preliminary scope will serve as the starting point for discussion at the scoping meeting. Arriving at the meeting with a clear, proposed scope, rather than relying on others to define it, ensures a more productive and focused discussion.

For projects involving consultants, it is important to meet with district staff early to finalize the scope before submitting a contract request through PEPS (Professional Engineering Procurement Services). The scope should be reviewed and approved by the Area Engineer, Design Engineer, Surveyor, and Environmental Team prior to submitting the contract request. This ensures that the scope is aligned with the project's objectives, environmental considerations, and budget constraints. Finalizing the scope internally helps to avoid delays, as the contract clock begins once the request is submitted.

In summary, effective project planning starts with a thorough site visit and scope development prior to the scoping meeting. When utilizing consultants to develop projects, members of the TxDOT team should visit the project site together. The team should include the project manager, environmental staff, area office personnel, maintenance section staff, or county staff responsible for off-system projects. By identifying key issues and proposing a preliminary scope, the project discussion can move forward efficiently, ensuring the project proceeds with a clear, well-defined direction.

1.5 Pre-Bid Contact and Approver

The Design Manager is responsible for updating the Project Managers tab in TxDOT Connect to include the Pre-Bid Contact and Pre-Bid Approver for each project.

1. Pre-Bid Contact

- For the Pre-Bid Contact, use the two names listed in the Yoakum District Master General Notes under the General Section. These names are designated as the primary contacts for all pre-bid inquiries.

2. Pre-Bid Approver

- For Construction Projects: The pre-bid approvers will be the Director of Construction at Yoakum District Office.
- For Maintenance Projects: The pre-bid approvers will be Director of Operations at Yoakum District Office.

3. Adding New Contacts

- If new names are added to the Yoakum District Master General Notes, these names must be reviewed and approved. To add new contacts to TxDOT Connect, submit a ticket for approval before including them in the system.

1.6 General Plan Sheet Preparation Guide

1.6.1 Preliminary Plans Preparation

1.6.1.1 Agreement

Prior to project letting, all necessary agreements or amendments must be completed and referenced on the title sheet of the plans.

It is essential to coordinate with District Maintenance and Traffic Operations regarding any existing maintenance agreements that may be impacted or require amendment due to the project. Relevant agreements may include landscaping, aesthetics, painting, lighting, and signals.

Landscape Maintenance Agreement (LMA): Any landscaping or aesthetic elements included in the project plans will require a Landscape Maintenance Agreement (LMA) or an amendment to an existing LMA. This agreement is mandatory for both on-system and off-system projects where landscaping or aesthetic features are involved.

Lighting Agreement: If the lighting is to be maintained by an external entity, a Lighting Agreement must be established. This agreement should be coordinated with the District Traffic Operations office.

1.6.1.2 Preliminary Online Posting District Guidance

- FTP Online posting of 90% Plans, Cross Sections, Construction Schedule, and Estimated Quantity at 5 months prior to Letting.
- FTP Online posting of 100% Cross Sections, Construction Schedule, and Estimated Quantity at 2 months prior to Letting.
- FTP Online posting of Final Cross Sections, Construction Schedule, As Built, and Geotechnical Report, Hydraulic Report 1 month prior to Letting.
- Plan Distribution:
 - ❖ Notification to the District Engineer, Director of Transportation Planning & Development, Area Engineers, and project managers are sent via email once the projects are posted online to the FTP site.
- Specific Guidance for online posting can be found under:
 - ❖ T:\YKMCONST\FTP\SOP Guidance

1.6.1.3 Standards

1. Check ProjectWise or the website for Yoakum District Standards

- ProjectWise Location: *YKM\Standards\District Design Standards\Approved* for approved design standards

2. Yoakum Non-Standard Details:

- *YKM\Standards\District Design Standards\Non-Standard Details*** for common details that require PE seal.

3. Work Zone and Barricade Standards

- Include all BC (Barricade and Construction) and WZ (Work Zone) standards in all plans. This includes the current rumble stripe standard (WZ(RS)-XX).

4. Traffic Control Plan Standards

- Include TCP (3-1)-XX through TCP (3-4)-XX standards in all on-system plans including roadway work. These standards are used for the installation and removal of lane closures.

1.7 Safety Score Tool

Each project must undergo an evaluation to assess the need for the application of one of the available safety scoring tools. This evaluation is necessary to determine whether the project requires the use of a specific tool based on the project's location, type, and potential safety concerns. Detailed information and access to the safety scoring tools can be found at the following link: [Safety Scoring Tools for Rural Highways](#).

General Knowledge

2.1 Survey

For geometric projects involving earthwork, new locations, widenings, including increased pavement width, added shoulders and lanes, and Super 2 projects, ADA/sidewalk improvements, and bridge replacements, a survey is required. However, survey work is not necessary for sealcoats or striping projects. For mill/inlay and overlay projects, a survey may be needed at the culverts to check for extensions. It is essential to check the accuracy of the survey and to ensure that the survey control sheets are included in the plans.

2.2 Stationing

For overlays, on-system bridges, rehabilitations, and small projects such as sidewalks and ADA improvements, it is important to match the existing stationing. For rebuilds, restationing is recommended, ensuring that the stationing increases in the same direction as the as-built plans. Provide a station equation at the beginning or end of the new roadway to tie the new project to the existing stationing. Additionally, it is preferred to avoid having the north point facing downward on a sheet, for example.

For On-System projects, stationing should follow the preferred order, starting with stationing from west to east or south to north. If this is not feasible, stationing can follow the direction of increase in reference markers or, alternatively, it can adhere to the currently established stationing.

For Off-System Projects, stationing must always increase from west to east or south to north, consistent with standard stationing practices.

When the roadway has multiple parallel alignments, these alignments should be clearly differentiated in the plans. For instance, the stationing for main lanes may be designated as “3755+00,” the northbound frontage road as “13755+00,” and the southbound frontage road as “23755+00.”

It is important to use alignment names that accurately describe the specific purpose of the roadway or feature. For example, the northbound frontage road should be labeled as "Northbound Frontage Road (NBFR)," the southbound main lane as "Southbound Main Lane (SBML)," the

northbound exit ramp as "Northbound Exit Ramp (NBEX)," and the southbound entrance ramp as "Southbound Entrance Ramp (SBEN)."

All alignments in the plans must reference the main lane to ensure clarity and consistency in the stationing system.

2.3 Design Speed

The design speed should meet or exceed the posted speed. Horizontal design speeds shall match the posted speed. Existing constraints may require a lower design speed for the vertical alignment, but it must not be lower than the minimum required in the Road Design Manual (RDM).

2.4 Clear Zone

The clear zone varies depending on the RDM and the road classification. Typically, the clear zone is 10' 16', or 30' in rural areas depending on traffic volume and 2R, 3R, or 4R design criteria, and 4 feet from the face of the curb in urban areas. Any existing infrastructure such as power poles, cabinets, or signal poles must be located at least 1.5 feet from the curb face in urban areas.

2.5 3D Model & Cross-Sections

A 3D model is required for geometric projects that involve cross-sections. All areas should be designed with precise attention to detail, ensuring that everything aligns correctly. A 3D model is not required for overlay, sealcoat, striping, SET, or culvert widening/replacement-only projects. It is advisable to first draw a 2D base file and then create a separate 3D corridor model to avoid accidental modifications by others who may not be familiar with the model.

When preparing the 3D model, it is essential to verify that the right-of-way (ROW) is drawn accurately, and drainage easements are included. Design side slopes as flat as possible. For typical 2-lane roads, the front slope should be 6:1 and the back slope should be 4:1 when buying Right of Way. When widening inside existing ROW, 4:1 front slopes are the usual max slope with 4:1 back slopes. These slopes can be increased when room doesn't allow near drainage structures or other constraints. It is preferred to steepen the back slope before steepening the front slope to 3:1.

For 4-lane divided highways, use 6:1 for the front and median slopes, and 4:1 for the back slopes. Front slopes may be increased when fills exceed 12' from the edge of pavement to the flowline of the ditch or front slope catch point with existing ground. Median slopes may increase up to 4:1 when the opposing lanes of travel have separate profile grade lines when the roadway is traversing the side of a hill. It is best to keep the front slope constant (6:1, for example) and maintain a constant distance from the centerline to the ditch bottom, letting the back slope tie into the natural ground at the ROW line. The invert of the ditch should be at least 6 inches below the subgrade crown. In all cases, flat bottom ditches are preferred over V-bottom ditches whenever room allows.

It is also necessary to match existing roadway and ditch elevations at the project limits. Cross-slopes should be checked to ensure proper drainage, and if necessary, the ditches beyond the project limits should be regraded to allow for proper drainage. Special attention must be paid to driveway cross-sections, particularly where culverts are involved, to ensure ditches are deep enough to accommodate an adequate structure height under the driveway and to provide enough cover over the driveway structure. If ditch flow lines are set too high, driveway profiles can have an undesirable profile over the driveway culvert. If the culvert flowline elevations are not properly set during design, a bump can form during construction. Additionally, additional cross-sections should be cut at tapers, driveways, and culverts.

2.6 Alignments and Grades

It is essential to differentiate multiple alignments, such as main lanes, frontage roads, and intersections. Horizontal deflections should be limited to 30 minutes or less for low-speed areas and 15 minutes or less for high-speed areas. Vertical grade breaks of 1% or less are acceptable for low-speed designs, while grade breaks of 0.5% or less are recommended for high-speed designs. Low speed is equal or less than 45 mph design speed. The minimum grade for curb and gutter is 0.35%. For 3R projects, horizontal and vertical check sheets are required. The recommended minimum length for vertical curvature should be three times the design speed, with values rounded to the nearest 5 feet. K values, as outlined in the RDM, should be referenced for vertical curvature design.

2.7 Vertical Alignment

When widening a roadway, it is desirable to maintain the existing profile unless otherwise required. Provide profile data to verify that the roadway aligns with the specified design criteria. The profile data may be presented in a table format.

However, the profile data is a guide and for design verification purposes only. Construct the pavement in accordance with the typical section.

2.8 Horizontal Alignment

Similarly, when widening or overlaying a roadway, the horizontal alignment should remain unchanged unless specific modifications are necessary. Provide alignment data to verify the roadway meets these requirements, which can also be presented in a table format.

However, the horizontal data is a guide and for design verification purposes only. Construct the pavement in accordance with the typical section.

2.9 Utilities

For utilities, the project begins by sending out a Notice of Proposed Construction (NOPC) to relevant utilities. All utilities, including municipal and TxDOT-owned ones, must be located and shown in the plan view. If conflicts are found, a Notice of Required Accommodation (NORA) is sent.

It would be important to emphasize that any design changes made after NORA (Notice of Receipt and Acknowledgment) is sent would become reimbursable. This ensures that all parties are aware of the cost implications and that proper documentation and approvals are in place for any adjustments to the design after the NORA stage. Clear communication about this policy can help avoid misunderstandings and ensure compliance with the project's financial and contractual guidelines.

It is important to design around existing utilities if feasible. Underground utilities should be potholed, and the tops of lines should be surveyed to confirm their location. Close coordination with the utility coordinator and utility companies is essential. Utilities need to be clear at least five months before the letting date. If utilities will not be clear by this deadline, a Construction Management Plan (CMP) may be required. Any conflicts with overhead power lines need to be addressed early in the design phase. Special care should be taken to mark areas where utilities like

shallow waterlines or clay pipes may be vulnerable, especially when using heavy machinery like vibratory rollers. Additional considerations, like overhead power line clearance for light poles, should be factored into the design.

2.10 Schematics

Schematics are required for new location projects, projects requiring ROW acquisition, and those with added capacity. Schematics need to be reviewed by the Design Division and approved by the District Engineer. The environmental process should begin once the schematic reaches 30% completion. Control of access needs to be considered for interstates, and these plans should be shown clearly on the schematic. When designing driveways, special attention must be paid to access management, especially when denying access to specific areas like within 425 feet of an intersection. The project should ensure that structures, including bridges, are designed for future widening, and it's best to plan the bridge to ultimate width to avoid costly adjustments in the future.

2.11 Pavement Design

For pavement design, pavement coring and analysis must be completed early in the project to allow for proper estimation of materials. When increasing the thickness of layers for asphalt or concrete, it's crucial to ensure that the design life meets project requirements (typically 20 years for asphalt and 30 years for concrete). If necessary, areas with poor base material may require lime/cement treatment to ensure adequate compaction. Hot mix design should focus on ensuring that each layer bonds properly to the layer beneath it, and tack coats are necessary to maintain this bond.

2.12 Typical Sections

For typical sections, it's important to keep them simple and focused on the most common conditions. Each subsequent layer in a pavement design should be, at least, 6 inches wider than the previous one to ensure stability. Typical sections should show and label the pavement layers, thickness and widths, lane and shoulder widths, pavement edge tapers, front slope and back slope rates, ditch bottom with, and right of way width. This applies to both existing and proposed typical sections.

2.13 Phasing & Traffic Control

When developing phasing and traffic control plans, it's critical to ensure that the construction process is clear and feasible for the contractor. Phasing should be simple and easy to understand, with special care taken in urban projects to minimize disruption to local businesses and residents. Temporary striping, barriers, and other elements must be provided, especially if traffic is expected to be redirected for extended periods of time. The plans should also consider whether temporary signals or illumination will be necessary, as well as specific measures for handling water-filled barriers, detours for beam placements, or issues with railroad crossings.

Please note that only state roads should be used for detour routes. Detours on non-state roads may not meet necessary safety and regulatory standards, and as such, are not approved for use unless otherwise specified and approved by the relevant authorities.

2.14 Overlays/Mill & Inlays

For overlays and mill-and-inlay projects, it is essential to check structures for any needed extensions and estimate 2-5% surface area for pavement repairs, especially if the existing pavement is in poor condition. Overlay projects must consider milling the project ends to avoid bumps. Special care must be taken when dealing with existing pavement issues such as broken shoulder slopes or soft spots that may need repairs.

2.15 Super2 Projects

The Super 2 project involves widening the roadway on both sides to increase traffic capacity and improve safety. In addition to the widening, the project will include extending existing culverts to accommodate the increased water flow resulting from the expanded road. This will ensure proper drainage and prevent potential flooding or erosion issues, ultimately enhancing the long-term functionality and durability of the roadway.

2.16 Districtwide Sealcoats

Seal Coat season is from May 1st to September 15th. Use Type II striping and Type C prefab markings, omitting retroreflective requirements.

2.17 Bicycles & Pedestrian

If the cost of pedestrian elements exceeds \$50,000, a review by the Texas Department of Licensing and Regulation (TDLR) is required. Sidewalks should be designed with a minimum width of 6 feet at the back of the curb and 5 feet in other areas. The design must adhere to the Public Rights-of-Way Accessibility Guidelines (PROWAG) standards. Driveways that do not meet the required cross-slope specifications must be replaced. A minimum clearance of 48 inches should be maintained around all obstacles, and any items that do not meet this requirement must be relocated. Work outside of the Right-of-Way (ROW) should not be shown unless a signed Temporary Construction License is obtained. Sidewalks should be designed with a 1.5% cross slope. Ramps should be paid for based on square yardage (SY). If historic structures are involved, design considerations must be made in accordance with applicable preservation standards.

2.18 Hydrology and Hydraulics

The design storm must be clearly defined, ensuring that the 100-year flood elevation does not exceed limits that could impact adjacent properties. Both existing and proposed conditions should be analyzed to design the roadway profile in accordance with the high-water elevation of the design storm. While the 100-year event may overtop, excessive depth should be avoided. For off-system bridges, the design standard should meet or improve as the existing structure. Recommended hydraulic software includes HY-8 for culverts, ORD-DU for storm drains, and HEC-RAS for bridges. The Kirby-Kirpich method is recommended for calculating Time of Concentration. Pipe or box velocities should be kept between 3 fps and 8 fps to prevent downstream erosion. A drainage analysis is required when extending culverts by a total of 8' or more. A scour analysis is necessary for new bridges over water, with the design flood elevation being below the lowest beam. Consider the need for relief culverts and evaluate the potential for stream meandering, which may require realignment or bank stabilization. Coordination with the local floodplain administrator is essential throughout the design process and must be documented on the Hydraulic Data Sheet in the plans.

Bridge-class structures must be properly protected, not simply extended past the clear zone. Adequate safety measures, such as barriers or guardrails, should be implemented to prevent vehicles from impacting the structure and to ensure the safety of all road users.

2.18.1 Culverts

Field verification of culvert types and sizes is crucial; reliance on as-builts or survey data alone is not recommended. Assess the condition of all culverts to determine if they need replacement or repair, and measure breakback at pipe joints. When constructing new culverts, aim for consistent side slopes, as mixing slopes can create construction challenges. Parallel pipes also require slopes. Flared wings are preferred over straight wings for crossing structures to improve hydraulic performance. Headwalls should typically be 3" tall to meet clear zone requirements and broken-back culverts are acceptable if they prevent significant drop-off protection. Consider inverting the inlet side or using drop inlets in areas with limited ROW or steep slopes. Always prioritize safety end treatments (SETs), and use cast-in-place SETs. Design should ensure adequate backfill around culverts and storm drains and verify the compatibility of old and new pipe designs when extending structures.

2.18.2 Storm Drains

Storm sewer systems must be designed for gravity flow to ensure that the hydraulic grade line (HGL) remains below the top of pipes or boxes. Ensure that the HGL does not exceed the elevation of any inlets or manholes, which would lead to surcharge conditions. The soffit elevations of storm sewers should be properly matched across the system. Avoid inverted siphons and consider relocating utilities if necessary to maintain proper flow conditions. Proposed curb inlets should be evaluated for their ability to fit within the ROW while avoiding conflicts with other utilities. In sag conditions, where water cannot naturally drain, three inlets should be used to ensure proper drainage. Design all grates for durability, especially those located in areas subject to snow plows. The preference is for PCO curb inlets, though PCU inlets may be necessary in areas with more restrictive ROW. Also, assess whether alternative materials like HDPE are needed for pipe systems instead of traditional reinforced concrete (RC) pipe.

2.18.3 Riprap

Riprap is critical in preventing erosion, particularly at sawtooth curb openings, where it should extend from the gutter to the ditch. A riprap flume should be installed at

the transition from curb and gutter to a ditch. When large volumes of water are expected, deeper toe-walls are necessary for stabilization. Ensure that caverns under existing riprap are addressed by adding cement-stabilized backfill. Riprap should also be placed in areas that are difficult to mow; if the space between retaining walls and MBGF is tight, concrete riprap may be a better solution to avoid maintenance challenges. For bridge header riprap, consider intermediate toe-walls to counteract gravity and recommend weep holes to address hydraulic pressure on large riprap areas. Slopes steeper than 4:1 should be riprapped to reduce maintenance difficulty. Avoid using stone riprap unless there's a compelling reason, such as flowing river conditions, as it may require more maintenance and is less effective in some environments.

2.18.4 Gabions

Gabions are structures designed to provide erosion control and support in areas prone to water flow. When designing gabion systems, it is crucial to ensure they are wide enough to prevent erosion. It is insufficient to place gabion mattresses only at the bottom of the streambed; these structures should extend up the embankments and ditches for optimal effectiveness. Additionally, the incorporation of a toe-wall may be necessary in some cases to enhance stability and prevent undermining of the gabion structure.

Gabion mattresses should not be installed on steep slopes, as this application is not ideal for their intended function. In such cases, alternatives such as stacking gabion baskets in a stair-step fashion, or using concrete riprap with velocity dissipators, may be more appropriate. Furthermore, to improve the stability of the gabion mattresses, contractors may need to cover them with soil and grass, providing added support and reducing the risk of erosion.

2.18.5 Retaining Walls

Retaining walls are designed to support the soil and prevent erosion or collapse on sloped terrains, especially when managing changes in grade. In the case of asphalt pavement sections, a moment slab is required, and the use of the RW(TRF) standard should be specified. The design should clearly define the pattern of the retaining wall and include provisions for managing water flow at the bottom of the wall. A drop inlet is often preferred

in these instances to prevent water from accumulating on the frontage road, where it could freeze during winter months.

To ensure proper design and performance, it is essential to request retaining wall borings. The number and location of these borings must adhere to the specifications outlined in the Geotechnical Manual. Underdrains are also a critical component of retaining walls, and these should be incorporated into the design. Underdrains must be detailed in the wall profiles, and the system must include an appropriate outfall to direct water away from the structure.

When placing embankment or select fill around the wall, it is important to do so in 8-inch lifts. This prevents the formation of vertical joints between regular fill material and select fill, promoting uniformity and stability in the final structure. Earthwork payments should be calculated based on the dimensions from the outside of the coping to the outside of the coping, and the wall should be embedded a minimum of 2 feet below grade to ensure stability.

Retaining walls may also feature artistic elements, and this design should be specified at the outset. Compaction of the earthwork must be performed from the outside in and from the bottom up to ensure the structural integrity of the wall. During construction, if there is a need to bridge over utilities or culverts, the design must account for these features.

In cases of tall retaining walls, particularly those built over railroad tracks, the wall straps may need to be longer as the height of the wall increases. Additionally, in freight corridors where super-heavy loads are expected, the design may require the extension of the wall straps, especially if a hot-mix section or moment slabs are involved. For areas using glass paving, the paving should be extended to the rail to act as a water barrier.

Finally, when determining the dimensions of the coping, the choice of material—either precast or cast-in-place—should be made early, as mixing materials may lead to inconsistent aesthetic outcomes. MSE walls are typically braced at a 1:1 ratio during construction, requiring adequate lateral space for proper installation of the braces. This space should be factored into the traffic control plan, as it is essential for both safety and operational efficiency. In addition, consideration should be given to the accessibility of

bearing pads and beam ends for maintenance or repairs, particularly when the top of the coping is set too close to the beam bottoms.

2.19 Bridges

2.19.1 New Bridges / Bridge Replacements

When designing new bridges or replacing existing ones, certain critical design standards must be followed. The minimum clearance for most bridges is 16.5 feet, with a higher clearance of 18.5 feet required for freight corridors. This ensures that larger vehicles and equipment can safely pass beneath the bridge.

Additionally, abutments and bents must align with the slope of the bridge deck to maintain consistency in the structure. This alignment helps prevent issues that have been encountered in previous projects, such as improper drainage, uneven load distribution, or structural stress caused by misalignment. In some past projects, these inconsistencies led to accelerated wear and tear on the bridge, difficulty in road maintenance, or, in extreme cases, required costly retrofits to correct the alignment. By ensuring proper alignment of abutments and bents, such issues can be avoided, resulting in a more durable and efficient bridge design.

In terms of environmental considerations, bridges must undergo inspections to check for lead and asbestos. If these materials are found, mitigation plans need to be developed. Bridges crossing railroad tracks may also require additional design features, such as deflection walls at the railroad right-of-way (ROW). For bents, 3-column configurations are preferred, as they provide better structural integrity, while 2-column bents necessitate more extensive protection due to their potential for catastrophic failure if struck. Where feasible, spans should be designed to clear the railroad ROW.

Bridge length should be sufficient to avoid sight distance issues caused by retaining walls. Additionally, when spanning streams, the mitigation cost of placing a box culvert may exceed that of a bridge, particularly if the box culvert already exists. When it comes to materials, steel beams require permanent metal deck forms, and the overall bridge layout must accommodate all necessary details such as fixed ends, dowels, and shear keys. Special

attention should be given to the distance between the beam bottom and the 100-year water surface elevation, especially when shear keys are required.

2.19.2 Bridge Repairs / BMIPs

Bridge repairs or Bridge Maintenance and Improvement Projects (BMIPs) require a tailored approach to ensure the longevity and safety of existing structures. For bridges with hot-mix overlays, these should be replaced with multipolymer or polyester overlays, which provide better durability. Repairs to bridge columns should be done carefully, with no more than one-third of the column's height and circumference repaired at a time. This prevents potential issues such as column buckling, which can compromise the integrity of the entire structure.

When replacing header joints, payment should be based on volume, and dimensions must be carefully verified. If the diaphragm at the abutments is attached to the backwall, it is crucial to address this issue, as the bridge cannot be lifted for repairs if the diaphragm remains connected. Additionally, when performing full-depth or partial-depth repairs, it is important to inspect the deck for signs of further damage. Sounding the deck can reveal areas that require additional repair, and after milling, it is often found that more areas need attention than initially anticipated.

In some cases, embedded galvanic anodes may be required for repairs. These anodes spaced every 18 inches around the perimeter of the repair area, help prevent future corrosion, adding an extra layer of protection to the structure. This type of repair is typically outlined in special specifications.

2.19.3 Bridge/Wall Painting, Staining, or Repainting

The maintenance of bridge and wall surfaces through painting, staining, or repainting requires a specific process to ensure the durability and aesthetic quality of the structure. When applying anti-graffiti coatings, it is essential to ensure compatibility with the underlying paint or stain. The wrong combination of coating and paint can lead to fading or deterioration over time, as seen in previous projects where improper coatings caused issues such as fading in the sun.

The application process begins with water blasting to clean the surface, followed by the application of paint or stain. Once this layer is fully dried, an anti-graffiti coating should be applied. The anti-graffiti coatings come in different types; Type III coatings, for example, are sticky when dry, and they tend to accumulate dust, which can detract from the bridge's appearance.

Additionally, when using anti-graffiti coatings, any reflectors or delineators must be attached with fasteners, as adhesives will not adhere to the coating. This ensures that all additional features, such as reflectors or safety markers, remain securely in place, even in challenging weather conditions. Special attention should also be given to the environmental factors, as improper application can result in more frequent maintenance and repair needs.

2.19 Signals

When designing signals, it's essential to provide both existing and proposed layouts. The layout should include utilities for reference. Ensuring that the drill shafts do not interfere with drainage, utilities, or other infrastructure is critical. The correct diameter for drill shafts should be drawn, ensuring they fit within the Right of Way (ROW), and the XY coordinates or station and offsets for the drill shafts should be included. Signal poles must be positioned at least 4 feet from the curb face to avoid interference with vehicles and pedestrian movements.

In terms of components, ensuring that audible push buttons, countdown pedestrian heads, and proper signage are included in the design is mandatory. Pedestrian buttons must be placed within 10 feet of ramps to ensure accessibility. Also, consider how any signs mounted on the mast arms will be paid for, as this can sometimes require separate itemization.

2.20 Illumination

For illumination designs, layouts are required, including references to utilities for coordination. Each assembly should be identified with a circuit letter and run number (e.g., A-2). LED lights should be used for all fixtures to ensure energy efficiency and long-term reliability. For interchanges or rural intersections where right-of-way allows, high-mast lighting should be considered. It's important to coordinate with maintenance directors to ensure that maintenance

personnel have the proper equipment, like a bucket truck, to maintain poles, especially those 50 feet or taller.

Underpass lighting is recommended when continuous lighting exists on both sides of the bridge, enhancing visibility and safety. Additionally, the voltage drop for each circuit should be calculated using NEWVOLT software, and photometric cells should be activated by the 30% plan submittal to review lighting coverage. For lighting installations, ensure that no more than 60 LED lights are placed per service and that pole locations do not conflict with overhead or underground utilities.

For poles within the clear zone, ensure they are placed on transformer bases to minimize hazards, and if the pole is outside the clear zone but near parking lots or businesses, non-breakaway poles should be used. Clarify which power company serves each side of the installation and coordinate any necessary power line rerouting. When installing underpass lighting, detailed drawings showing how the conduit mounts to the beams or bents must be provided to avoid confusion during construction.

2.21 ITS (Intelligent Transportation Systems)

For ITS designs, a comprehensive layout is needed, including camera placement, message boards, and conduit runs. The District Traffic Section should be consulted for guidance on these placements. Concrete ground boxes with steel lids should be used, as polymer ground boxes are not acceptable. When mounting message boards, particularly on skewed angles, a detailed drawing for proper anchor bolt placement is necessary to avoid structural issues.

The ITS components should follow specific guidelines such as using SP000-395 for ITS items that aren't the major work item and SP002-007 for standalone ITS projects. It's important to ensure that all elements of the ITS system are clearly identified and coordinated in the plans to avoid delays during construction.

2.22 Signing

A comprehensive sign layout is an essential part of the transportation project design process, ensuring that all signs are clearly indicated with proper numbering and in compliance with relevant guidelines. In some cases, it may be permissible to combine the sign layout with the striping layout;

however, caution must be exercised to avoid overcrowding the plans, which can lead to confusion during construction and installation. The goal is to provide a clear, readable, and organized sign layout that aids in navigation and ensures safety for road users.

When designing new signs, the project should be treated as though there are no existing signs, ensuring that the design process follows the Sign Crew Field Book and the Texas Manual on Uniform Traffic Control Devices (TMUTCD) guidelines. These documents provide the standards and specifications for the proper design and placement of traffic control devices. It is critical to recognize that existing signs may be outdated, mislocated, or improperly designed. Therefore, verifying the position and condition of every sign before making design decisions is essential. However, there are exceptions—signs such as reference markers, speed limits, and city/county boundary signs cannot be relocated without special considerations.

For large guide signs and D-series signs, the use of SignCAD (Sign Computer Aided Design) is mandatory. SignCAD is a software tool that facilitates the accurate design and layout of complex signs, ensuring consistency and compliance with design standards. Additionally, it is important to address the placement of sign bridges. These structures must be strategically placed to serve both directions of travel and should be located outside the clear zone to minimize safety hazards. Proper placement of sign bridges ensures that they do not interfere with roadside recovery areas or create potential obstacles for errant vehicles.

For large signs or signs requiring sign bridges, cross-sections must be provided in the design plans. These cross-sections help to visualize the relationship between the sign, the roadway, and surrounding elements, ensuring that the sign placement is practical and meets safety standards. Moreover, the plans should clearly specify whether the contractor is responsible for salvaging, storing, and remounting any existing signs. This is an important detail to clarify to avoid misunderstandings and ensure that project timelines are met.

Finally, the visibility of sign bridges is a key consideration. Sign bridges must be positioned within the line of sight for drivers, ensuring that signs are legible and provide adequate advance warning. Proper sight distance calculations and cross-section evaluations should be included in the design to guarantee that signs are both visible and effective in conveying critical information to motorists.

In summary, designing and laying out signage for a transportation project requires attention to detail and strict adherence to guidelines. Ensuring that signs are correctly numbered, properly

placed, and clearly visible helps to enhance safety and driver understanding. Additionally, incorporating modern design tools, such as SignCAD, and verifying the condition and placement of existing signs ensures that the final project is compliant with all safety standards and regulations.

2.23 Striping, RPMs, and Delineation

A layout of striping and delineation is required, and can often be combined with the signing layout if space allows. Simple overlays or sealcoats typically do not require a full layout but should include a summary. Use "puppy dog tracks" for dual left and right turns, with the outside lane turning into the farthest lane and the inside lane having the option of the other two.

Use Type II-C-R RPMs for 4-lane roads and frontage roads, and Type I-C RPMs for undivided roads and city streets. Red Type I-A RPMs should be used for wrong-way arrows on ramps. Ensure the appropriate striping for lane drops and for the ends of freeways. Delineators must also be used, and details should be included for median deceleration lanes, including object markers. Continental crosswalk striping should be used for pedestrian crossings. Avoid using "WORD" markings unless directed by the District Traffic Office.

2.24 Railroad

Effective coordination with railroads is a critical component of transportation project planning, especially when construction work is near railroad crossings. Early communication should begin if any portion of the project falls within 500 feet of a railroad crossing to ensure proper planning and safety measures are implemented.

When construction activities occur within the railroad Right-of-Way (ROW), it is necessary to establish a formal agreement with the railroad operator. This agreement may outline specific conditions and responsibilities related to the work within the ROW. Additionally, Scope of Work sheets must be included in the project documentation for any work, including TCP that occurs within 500 feet of the crossing to clearly define the scope and extent of the proposed activities. In some instances, Right-of-Entry permits, or railroad insurance may be required to address safety and liability concerns.

For new at-grade crossings, it is important to adhere to the guidelines provided by the AASHTO Green Book and the Railroad Design Manual (RDM). These guidelines outline the

specific design standards for such crossings, particularly concerning approach grading. The approach grading must ensure that the roadway entering the crossing includes a 0% grade for a minimum of 50 feet on both sides of the railroad tracks to facilitate safe and smooth vehicle movement across the crossing.

Another key consideration is the sight distance triangle for railroad crossings, which must be evaluated to ensure adequate visibility for drivers approaching the tracks. This is especially important when the railroad tracks are located below grade and surrounded by berms, as these conditions may obscure the driver's line of sight. In these cases, it may be necessary to regrade the adjacent roadways to meet the required sight distance standards, ensuring that drivers have clear visibility of approaching trains and can stop or proceed safely.

In summary, coordination with the railroad must be initiated early in the project development process, particularly if construction occurs near or within the railroad ROW. It is essential to adhere to the minimum vertical clearance requirements for railroad overpasses to ensure safe passage of trains. Additionally, compliance with relevant design standards, safety requirements, and sight distance considerations is vital to ensure the safe and efficient operation of both the roadway and railroad systems.

2.25 SWP3 and Environmental

Stormwater Pollution Prevention Plan (SWP3) narrative sheets are required for the majority of the projects, with layouts included for projects covering an area of one acre or more. An NOI (Notice of Intent) is required for projects covering five acres or more. Coordination with the District Environmental Coordinator is vital to address design, permits, endangered species, and historical concerns early on. Phased designs should be used to prevent contractors from installing all Best Management Practices (BMPs) at the start of the project. SWP3 details should include quantities for materials like logs, sandbags, and silt fences, and each BMP must be clearly labeled with installation and removal dates.

If there are any historical or endangered species concerns, ensure those are incorporated into the SWP3 as well. Urban projects typically involve topsoil, drill seeding, and hydro-mulch for vegetation, while rural projects often focus on blading back existing vegetation. For projects near Water of United States (WOTUS), ensure drill shafts are considered as fill, and be aware that any disturbance near critical habitat for wildlife may require modifications to the design.

For projects that disturb 10 acres or more, the project is considered an impacted project. As such, the SWP3 must be signed and sealed by a Professional Engineer (P.E.) or a Certified Professional in Erosion and Sediment Control (CPESC). This certification ensures that the stormwater management plan has been thoroughly reviewed and meets all applicable regulatory and engineering standards.

For ACP Overlay and Seal Coat projects, a SWP3 is not required, as these types of projects typically do not involve significant land disturbance or potential for erosion. However, for all other projects, compliance with SWP3 requirements is necessary.

2.26 Construction Schedule

The construction schedule is a critical tool for managing the timeline and sequence of activities throughout a project. It should be created using scheduling software such as P6 or Microsoft Project, which allows for tracking relationships between activities, start and finish dates, and important milestones. The schedule serves as a roadmap to ensure that the project is completed on time and within budget, while also highlighting potential risks or delays that could arise during the construction process.

Dates should only be shown for projects that have an established Construction Management Plan (CMP). The CMP provides a detailed framework for the project's execution, including critical timelines, resource management, and risk mitigation strategies. For projects without a CMP, the schedule should focus on key milestones and broad activity timelines without specifying exact start and finish dates, as these details may not be fully defined.

It is recommended to set aside five weather days per month in the construction schedule to account for potential weather-related delays. This weather contingency helps ensure that the project remains on track even if inclement weather causes interruptions. Additionally, extra time should be allocated for special activities that may require unique considerations, such as railroad work, SMA season (Surface Mixture Asphalt), and specific performance periods for tasks like striping, signal, and illumination work. These activities may require additional planning and time, especially if they are subject to seasonal or regulatory constraints.

The schedule must also reflect assumptions regarding the number of crews, production rates, weather conditions, and other influencing factors. These assumptions should be carefully considered and periodically reviewed to ensure the schedule remains realistic. Furthermore, time

suspensions—such as work restrictions due to road closures, environmental concerns, or regulatory limits—must be factored into the timeline. By accounting for these suspensions, the project schedule can more accurately reflect the potential for delays and avoid unexpected disruptions.

A key component of the construction schedule is identifying and maintaining the critical path. The critical path represents the sequence of tasks that determine the overall project duration. It is essential to ensure that no overruns or delays in critical activities, such as bridge deck repairs, affect this path. If critical activities experience delays, they will directly impact the project completion date. Therefore, careful attention should be given to ensuring these tasks stay on track.

As the project nears completion, activities such as curing concrete, seeding, punchlist items, and final cleanup should be scheduled for the end of the project. It is important to allocate enough time for these final tasks to ensure all project requirements are met before the final inspection. These end-of-project activities can be subject to weather conditions and other factors, so adequate time should be built into the schedule to address them properly.

Finally, the overall construction sequence must align with the proposed schedule. The sequence should ensure that work items are performed in a logical and coordinated order, with minimal risk of conflict between different work crews or phases. This alignment helps keep the project on track and prevents delays that could arise from poor coordination or inefficient sequencing.

2.27 TxDOTCONNECT

TxDOTCONNECT is used to manage project details, including limits, reference markers, and project descriptions. Ensure that the number of days and barricade months are consistent with the construction schedule and estimate. Barricade information should be listed with the time schedules or TCP details to ensure proper sequencing. The resources tab must be filled out correctly, and pedestrian-related items must be accurately coded. These pedestrian items should align with the Texas Department of Licensing and Regulation (TDLR) project number, ensuring compliance with state regulations.

In contrast, bridge items and other relevant project components should also be properly categorized under the estimate, ensuring accurate resource allocation and compliance with applicable standards. The resources tab should be filled out correctly to reflect both pedestrian and bridge-related categories.

2.28 Estimate

Prior to finalizing the estimate for letting, it is essential to thoroughly review the prices. Any new bid codes must be requested early in the process to mitigate the risk of last-minute delays. Quantities for roadway and bridge work should be clearly separated to ensure accurate cost allocation. Additionally, consideration must be given to the potential impact of late addendums or regional pricing adjustments on the final bid price. For bridge-related items, it is crucial to input the relevant bridge information under the Location tab to facilitate precise cost estimation. Contingencies should be set at higher levels early in the design process and gradually reduced as the project progresses. It is recommended to print the Estimate and Quantity (E&Q) directly from TxCONNECT, as the manual creation of a DGN file has been associated with errors. Upon finalization of the plans, the TxCONNECT estimate must be CAD sealed by the Project Engineer.

Traffic Control

3.1 Temporary Roadway Illumination

Temporary illumination is necessary if existing lighting is present along the corridor during construction. This must be maintained and addressed in the traffic phasing plan. The goal is to establish permanent illumination as early as possible during the phasing process.

Multiple Phases: Traffic control plans should be structured in clear phases. Use "Phase" and "Step" terminology (avoid "Stage").

For example:

Phase 1 → Step 1, Step 2

Phase 2 → Step 1, Step 2

3.2 Traffic Control Plan (TCP) Design

The TCP must be detailed and specific:

- Avoid referencing standards; design the plan with proper layout and phasing details.
- Clearly define the alignment of detours, geometry, and signing, including specific stationing for signs and any detour transitions.
- Verify conditions in the field (e.g., pavement widths, shoulder conditions) rather than relying on old plans.
- If existing overhead signs are impacted, coordinate with the Traffic Section for temporary overhead signs.
- Cross sections for detour pavement should be included to address vertical challenges such as driveway tie-ins and transitions.

3.3 Multiple TCP Phases and Traffic Configurations

When a project involves multiple phases and traffic configurations, challenges often arise with the removal and replacement of striping. A common issue is that the removed striping can remain visible on the roadway, potentially confusing drivers. This is particularly problematic when transitioning from a conventional intersection to an innovative intersection design, as the old striping may not align with the new traffic flow.

To address this, it is recommended to propose a pay item. This method helps ensure a smoother transition and reduces driver confusion.

3.4 Overlays

When planning for overlay projects, there are multiple factors that must be considered to ensure proper execution and compliance with standards.

An initial site visit is crucial to identify areas in need of improvement, such as issues with bridges, pavement conditions, signage, lighting, or traffic signals. It is also essential to check with the Area Office (AO) and Maintenance Section for any outstanding issues that should be addressed before proceeding.

For overlay-only projects, include Item 134 for Type A Backfill, except in cases where the roadway is curbed. Base repairs should be coordinated with the Pavement Engineer, with the necessary quantities determined by non-destructive pavement testing methods like Ground Penetrating Radar (GPR). If there are traffic elements, such as signals or illumination, within the project limits, work closely with the District Traffic Section to determine what replacements or upgrades are necessary.

For Metal Beam Guard Fence (MBGF), Item 540 should be included to bring the fence up to new standards. This includes bid items for rail adjustments and potential height increases if overlaying is conducted adjacent to the existing guard fence. The application of shoulder texture may also be required.

When overlaying, avoid working on bridges without existing overlays or on direct traffic concrete bridge decks. For bridges within the project limits, mill off existing hot mix asphalt (HMA) before applying the overlay. Specifically, for 2-inch overlays, polymer nosing material at bridge joints, along with Item 438-6004 for joint sealant and cleaning. Coordination with the District Bridge Section is vital for follow-up actions concerning bridges.

If concerns arise regarding the vertical clearance of existing bridges, milling may not be a viable option, and coordination with the District Bridge Section may be necessary to assess any required changes in vertical clearance. In cases where adjustments are made, vertical clearance measurements will be required, and appropriate signage, such as a vertical clearance sign or bracket, should be incorporated. Special consideration must be given to avoiding any increase in the roadway profile within floodplains, as such changes could affect the Base Flood Elevation (BFE) evaluation. Additionally, a thorough review of the current striping layout is essential to

ensure that any updates conform to current standards, enhance traffic flow, and, where applicable, accommodate bicycle facilities.

3.5 Bridges and Bridge Safety

When working on bridges, safety and structural integrity are paramount. The first step in bridge-related work is reviewing the Lead and Asbestos report for the existing structures. If lead or asbestos are present, coordinate with the District Bridge Section and District Maintenance for the necessary remediation plan before the project is let.

If the project involves the removal of a steel superstructure, ensure the prime contractor provides a demolition plan that details where and how the steel members will be cut for salvage. Salvaging lengths should be coordinated with the District Bridge Section and included in the design documents. If repainting a superstructure with lead-containing paint, TxDOT Item 446 addresses the mitigation process, which should be handled as part of the field cleaning and painting process, not prior to letting.

In terms of new bridge designs, it is essential to account for a 2-inch Hot Mix Asphalt (HMAC) overlay loading on all bridge structures. Should any issues arise with this additional load, coordinate with the District Bridge Section for further evaluation.

Bridge decks that do not currently have an overlay shall not receive one unless explicitly approved by the Director of Operations or Maintenance Engineer and the District Bridge Engineer. If an overlay is approved, it is essential to first seal the deck to prevent water infiltration. For sealing, a two-course surface treatment may be used for off-system projects, while on-system projects should utilize trackless tack with high-quality asphalt.

When reconstructing a bridge, avoid placing new drilled shafts too close to the existing foundation—ensure a minimum spacing of 2 shaft diameters between new and existing shafts. Existing drilled shafts should be removed 2 feet below the existing grade, while the remaining foundation is left intact.

3.6 Pedestrian Infrastructure

For projects that include pedestrian infrastructure, a separate pedestrian summary sheet must be prepared, detailing all pedestrian elements. If the total cost for these pedestrian elements exceeds \$50,000, the project must be submitted for review by the Texas Department of Licensing

and Regulation (TDLR) prior to being let. This review is necessary to ensure compliance with accessibility standards. Additionally, a TDLR Registered Accessibility Specialist (RAS) number must be obtained and noted on the project title sheet.

If pedestrian elements cost more than \$50,000, a TDLR inspection will be required. To allow for timely feedback, it is ideal to submit the design at 90% so that any recommendations can be incorporated before the final design is submitted. Keep in mind that variances for pedestrian design elements should be sent to the Design Division to be forwarded to the TDLR for approval.

TAMES (TxDOT Accessibility Management Enterprise System) outlines the process for incorporating pedestrian elements into roadway designs. The scope of work for the project must identify all necessary pedestrian infrastructure, including sidewalks, crosswalks, ramps, and safety measures. To ensure these elements are properly addressed, the report should be thoroughly reviewed to extract relevant pedestrian data, including existing conditions, deficiencies, and safety concerns. The report will also help assess the severity of pedestrian issues within the project area. If the roadway segment is located in a town or urban area, the Project Manager (PM) will coordinate with the district's Plan Reviewers to gather the necessary pedestrian-related information and ensure the design aligns with local regulations and standards. This collaboration ensures that pedestrian needs are fully considered in the design and that the project meets safety and accessibility requirements.

Ensuring that existing pedestrian paths are evaluated for compliance with the Americans with Disabilities Act (ADA) is compulsory. If the project will disrupt these paths, provide a temporary pedestrian route during construction. Design deviations or special design requirements should be coordinated with the Design Division for TDLR approval. If the project involves sidewalk retaining walls, use the appropriate construction methods.

Lastly, consider the impact of the construction on existing pedestrian facilities, especially those that do not meet ADA standards. An alternate route or temporary path must be provided to ensure continuous pedestrian access during construction activities.

Design Support

4.1 Preliminary Estimate

A preliminary estimate is an essential component of every project and must be prepared and stored for future reference. This estimate provides a baseline for project budgeting and can be used throughout the design and construction phases.

For projects that fall under Funding Category 8, preliminary estimates can be obtained by contacting the district traffic section. Once acquired, the estimate should be saved in the designated ProjectWise folder.

Preliminary estimates are also stored on the network at the following path: T:\CentralDesign\Estimates. These estimates are organized by counties and further categorized by control sections. The specific directory for each district includes counties such as Austin, Calhoun, Colorado, DeWitt, Fayette, Gonzales, Jackson, Lavaca, Matagorda, Victoria, and Wharton, among others. This structure helps to keep estimates organized and easily accessible for project teams working within these areas.

4.2 Utilities

Utility coordination is a critical aspect of project planning, and it is important to begin the process as early as possible to avoid delays and conflicts during construction. The process starts by sending out the Notice of Proposed Construction (NOPC) to utility providers, which informs them of the upcoming project and allows for the identification of any potential conflicts with existing utilities. It is also essential to identify and map all utilities—both municipal and TxDOT-owned—on the project plans, including those that are underground and overhead. This information is vital to determine if there are any conflicts with the proposed design. If conflicts are found, the Notice of Required Accommodation (NORA) should be sent to the utilities, requesting that necessary adjustments be made.

When possible, the design should be adjusted to avoid conflicts with existing utilities. However, in cases where conflicts are unavoidable, potholing of underground utilities is necessary to confirm their exact locations. This involves digging small test holes to expose the utility lines and allow for accurate surveying of their locations. This step is crucial for preventing utility damage during construction and refining the design to accommodate the utilities.

The utility coordination process also involves working closely with a utility coordinator to address any required agreements. All utilities must be cleared from the construction zone five months prior to project letting. If utilities are not cleared by this date, a triple zero provision should be included in the project to indicate that the utilities will be cleared after construction begins. If utilities will not be clear until three months or more after the letting, a Construction Management Plan (CMP) must be developed. Additionally, signed clearance certifications are required to confirm that all utilities are either clear or that any conflicts are identified along with the expected timeline for resolution.

To ensure compliance with all utility coordination requirements, preliminary utility certifications must be submitted at or before the 95% design submittal. District Transportation Design and Advanced Planning Section will request these certifications approximately five months prior to project letting, aligning with the time when the letting certification is requested by the Design Division. All utility coordination documents, including signed clearance certifications, should be stored in ProjectWise, the centralized project management system used to store and track project files.

Special attention must be paid to reimbursable utilities, as these require additional coordination. This process should begin as soon as the project is initiated to ensure that the necessary agreements and adjustments are made in time. Utilities that are reimbursable may involve more complex coordination and documentation, particularly if relocation or significant adjustments are required.

Additionally, it is important to check for conflicts with overhead power lines. For instance, when replacing or adding light poles, ensure that they maintain a 10-foot clearance from overhead power lines. If existing poles do not meet this requirement, adjustments in spacing or the pole selection may be necessary. In some cases, shallow waterlines, clay pipes, or other sensitive utilities may require the contractor to exercise extra care when working in these areas, such as avoiding the use of vibratory rollers, which could damage the infrastructure.

For widening projects, it is essential to carefully manage utility locates. A request for an 811 locate should be submitted for each culvert location, including GPS coordinates and a 200-foot radius on either side of the culvert to ensure a thorough utility investigation. To further assist in accurate utility locate results, a KMZ file containing the culvert locations can be submitted with the 811 request.

It is important to note that utilities within the railroad right-of-way (ROW) do not respond to OneCall. Specifically, utility accommodation and/or relocation requests within the Union Pacific Railroad (UPRR) ROW must be submitted directly to UPRR Real Estate through their online submittal system. The process for Canadian Pacific Kansas City (CPKC) may vary, so it is recommended to contact CPKC directly for clarification on their specific requirements and procedures regarding utility matters within their ROW.

Additionally, utility conflicts involving railroad utilities must be addressed. Be sure to check the Railroad Commission's website for information on any major utility lines located along the railroad corridor that may need to be avoided or relocated during construction.

4.3 Right of Way or Construction License Purchases

The acquisition of Right of Way (ROW) or construction easements is a crucial component of project development and must be carefully planned and coordinated throughout the project lifecycle. The need for ROW or easements should be discussed early in the process, specifically during the kick-off or Design Concept Conference (DCC) meeting. During this meeting, the timeline for ROW acquisitions should be established to ensure that the necessary easements and property acquisitions are completed well in advance of the project letting. This allows adequate time for any potential issues to be addressed before construction begins.

For projects involving ROW or construction easements, it is essential to complete the ROW lines early in the design process. The approval of these lines is a critical milestone, as they define the boundaries of the area required for construction or project activities. Once the ROW lines have been finalized and approved, the next step is to send the design file(s) (typically in dgn format) to the surveyor. The surveyor will then prepare the ROW maps and parcel plats, which are required for documenting the precise locations and dimensions of the property to be acquired. These documents are integral to the legal and administrative processes involved in securing the necessary ROW or easements.

By adhering to this timeline and involving the appropriate stakeholders early in the process, the project can ensure that the acquisition of ROW or construction easements is completed on schedule, minimizing delays and preventing complications during construction.

4.4 Plans Online for As-Built/Final Construction Plans

As-built plans, which document the final configuration of a project as it was constructed, are crucial for future reference and maintenance. To improve accessibility and streamline the process, many as-built plans are also available online. These plans can be accessed through the Apps link on the Crossroads homepage, which serves as a gateway to various project management tools and resources.

Once scanned, these plans are initially stored in ProjectWise, a document management system used by TxDOT. These plans remain in ProjectWise until the corresponding project is complete. After that, they will be added to the Plans Online website, where they can be accessed by staff and stakeholders.

4.5 USDA Soil Survey

The USDA Soil Survey provides valuable soil data and mapping for understanding soil characteristics across various regions. This tool can be accessed online through the Web Soil Survey portal at <https://websoilsurvey.nrcs.usda.gov/app/>. The soil data gathered from the survey helps in evaluating soil properties, including drainage, stability, and load-bearing capacity, which are essential for pavement design and project planning.

4.6 High Water Data

High water data is essential for assessing the potential flooding impacts on roadways. Maintenance sections that regularly experience high water over roadways provide valuable information for this analysis.

4.7 Road Closures

When road closures are necessary, clear communication and coordination are required. For projects requiring road closures, the specific closure details should be documented and communicated to the appropriate stakeholders to avoid any potential issues during construction. This ensures the public is informed and that traffic disruptions are minimized.

4.14 Permanent Traffic Count Sites

It is important to be aware of these traffic count sites when beginning a new project to ensure that the traffic data is considered in the design process. Coordinate with the District Traffic Engineer to identify specific locations and discuss relevant details.

4.15 Accident Data

Accident data, including wet surface crash data, can be requested by emailing the District Traffic Section Traffic Systems Administrator. For 2R projects, a crash analysis must be performed before the design reviews. The crash analysis ensures that safety considerations are incorporated into the project design. Additional accident data can be accessed via the provided KMZ file and Accident Analysis documents.

4.16 Traffic Data

Traffic data is essential for understanding traffic volume, patterns, and road performance. This data is vital for project design, especially for determining necessary improvements or adjustments to accommodate traffic flow and safety.

4.18 Superelevations

The maximum superelevation design should not exceed 6%. In superelevated sections, the algebraic difference in slopes between the roadway and ditch (or shoulder on divided highways) should not exceed 8%. For example, if there is a 4% superelevation, the outside ditch front slope should not exceed 12% (8.33:1). All superelevation station transitions should be shown on the Plan and Profile sheets, either in a chart or directly on the plan views, with the correct slope directions.

For projects that involve rebuilding or widening the roadway, existing superelevations must be checked for proper slope. If adjustments are needed, additional flexbase or HMA (Hot Mix Asphalt) level-up should be incorporated into the project design. A superelevation data sheet must be prepared to document slopes and any additional material required.

4.19 FEMA Coordination

All projects must be checked to determine if they cross a FEMA-mapped floodplain. If the project crosses a floodplain, a coordination letter must be sent to the county FEMA administrator along with a project map. The letter and map should be filed in the project folder, and a copy must be sent to the district ENV coordinator.

4.20 Railroad Coordination

Early coordination is essential for projects that cross a railroad. Specific plan sheets need to be included in the PS&E, and coordination should be done through email with the district railroad coordinator. It is advised to include the District Design Manager on all correspondence.

4.22 Stage Gate Checklist

A PS&E Stage Gate Checklist is required for all projects unless the district environmental coordinator has determined it is not needed. The checklist will be initiated by the environmental section and sent to the designer and/or project manager for completion. The signed checklist must be submitted with the supporting documents during design reviews and stored in ProjectWise.

4.23 3R Project Documentation

For all 3R (Resurfacing, Restoration, and Rehabilitation) projects, specific design information must be documented in the plan set, as outlined in the Roadway Design Manual, Chapter 4, Section 7.

4.24 Intersection Control Evaluation (ICE)

The Intersection Control Evaluation (ICE) process is required for preliminary or schematic design when certain stages (such as approved right-of-way footprint or 30% plans) have not been completed prior to June 30, 2025. Additional information can be found on the Design Guidance SharePoint or by contacting Design Division for support.

4.25 Intersections and Driveways

To ensure smoother driving conditions across intersections, the cross-slope approaches should be designed to a maximum of 1%. This will help prevent a rough ride when crossing intersections. In city sections, it is crucial to match existing gutter elevations as closely as possible to ensure proper drainage from the existing elevation at the ROW to the gutter. If transitioning to a single slope across intersections for drainage purposes, care must be taken to avoid creating low spots in the profile where water could collect. It is recommended to show curb and gutter grades every 20 feet on the profile to ensure correct drainage. It is also helpful to cut a profile at the ROW line during design to compare the proposed curb and gutter design with existing conditions. The ROW line tie-in point should be higher than the back-of-curb elevation to avoid causing flooding.

4.26 Sidewalk Design

Sidewalks should be designed with a 1% cross-slope, and it is important to avoid designing to the minimum allowable slope. Providing adequate space for field adjustments is essential.

4.27 Pavement Transitions

For pavement transitions, typical ratios such as 25:1 or 50:1 are commonly used. Alternatively, the formula $L = W \times S$, where W is the width being transitioned and S is the design speed, can be used to calculate the appropriate length for pavement transitions.

4.23 Alignments and Grades

It is essential to differentiate multiple alignments, such as main lanes, frontage roads, and intersections. Horizontal deflections should be limited to 30 minutes or less for low-speed areas and 15 minutes or less for high-speed areas. Vertical grade breaks of 1% or less are acceptable for low-speed designs, while grade breaks of 0.5% or less are recommended for high-speed designs. The minimum grade for curb and gutter is 0.35%, although it can be reduced to 0% as long as drainage is properly maintained. For 4R and 3R projects, horizontal and vertical check sheets are required. The recommended minimum length for vertical curvature should be three times the design speed, with values rounded to the nearest 100 feet. K values, as outlined in the RDM, should be referenced for vertical curvature design.

Plan Layout and Documentation Standards

5.1 Title Sheet

The Title Sheet for each set of plans is a critical component that provides essential information about the project. It should be prepared using the Yoakum District Title Sheet template (go-by). This ensures consistency and alignment with district standards.

5.1.1 Signature Line

The layout and signature blocks on the right side of the sheet should remain unchanged, as they contain standard information for signature and approval purposes. When off-system projects, the County Judge must sign the Title Sheet to provide concurrence.

INTRIM REVIEW DOCUMENT INCOMPLETE. NOT INTENDED FOR PERMIT, BIDDING, OR CONSTRUCTION. ENGINEER: <u>JohnPaul Clark</u> P.E. SERIAL NO: <u>123456</u> DATE: <u>11/07/2024</u>	
JPC Engineers San Antonio Austin Houston Dallas-Fort Worth 123, Main Road, Yoakum, TX 77995 Texas Engineering Firm #12345 Texas Surveying Firm #12345	
SUBMITTED FOR LETTING	
RECOMMENDED FOR LETTING	DESIGN ENGINEER
DIRECTOR OF TRANSPORTATION PLANNING AND DEVELOPMENT	APPROVED FOR LETTING DISTRICT ENGINEER
TEXAS DEPARTMENT OF TRANSPORTATION © 2024 BY TEXAS DEPARTMENT OF TRANSPORTATION ALL RIGHTS RESERVED	

5.1.2 Project Details

On the upper right corner of the title sheet, the following key project details should be included:

- **Function Class:** This defines the functional classification of the roadway, which will help categorize it for funding, design standards, and construction requirements.
- **Design Speed:** The design speed is determined based on the project scope, terrain, and roadway type, reflecting the speed at which the roadway is designed for safe travel.
- **ADT (Average Daily Traffic):** The ADT is used to assess traffic volumes and is essential for determining design features such as lane widths, shoulders, and other operational characteristics.

FED/STA DIVISION		PROJECT NO.		SHEET NO.
6		C 89-6-93		1
STATE	STATE SECT.	COUNTY		
TEXAS	YKM	WHARTON		
COUNTY	SECT.	JOB	HIGHWAY NO.	
0089	06	093	US 59	

HWY FUNCTIONAL CLASS: RURAL PRINCIPAL ARTERIAL
DESIGN SPEED: N/A
ADT: 28,600 VPD (2021)
41,184 VPD (2041)

PROJECT LENGTH	
ROADWAY	= 4,280.00 FT = 0.810 MI
BRIDGES	= 00.00 FT = 0.000 MI
TOTAL	= 4,280.00 FT = 0.810 MI

For projects with multiple answers for each of these categories (i.e., multiple CSJs or segments with different functional classifications, design speeds, or ADTs), the limits should be indicated clearly. These limits should ideally be shown by station limits or by CSJ number if the figures are consistent across the project segments.

5.1.3 Description of Work

The description of work must be formatted exactly as follows:

- The first part, represented by XXXX, must match the classification description shown in the Project Classification document for the project's Project Class Code as displayed in the TxDOTCONNECT (TxDOT Classification System).
- The second part, YYYY, must match the Project Description provided on the details page in TxDOTCONNECT.

Project Classification: RES - Restoration Short Description: Surfacing/Roadway Restoration Project Description: Restore Roadway

This standardization ensures that the description of the project is clear, concise, and aligns with official documentation in TxDOTCONNECT, minimizing errors or inconsistencies.

STATE OF TEXAS TEXAS DEPARTMENT OF TRANSPORTATION

PLANS OF PROPOSED STATE HIGHWAY IMPROVEMENT

FOR THE CONSTRUCTION OF A HIGHWAY SAFETY IMPROVEMENT PROJECT
CONSISTING OF CROSSOVER IMPROVEMENTS

US 59

WHARTON COUNTY

CSJ: 0089-06-093

PROJECT NO.: C 89-6-93

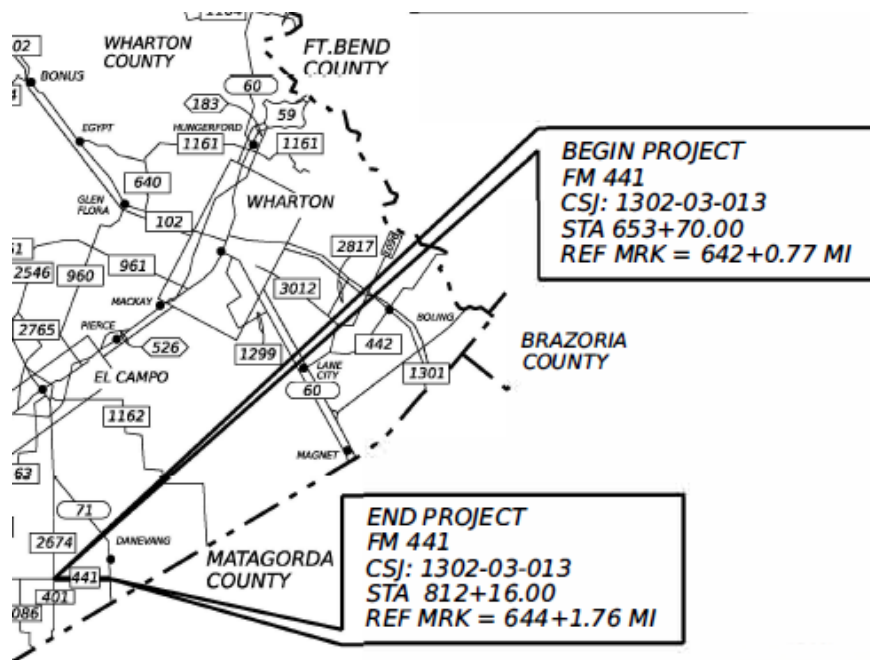
LIMITS: FROM 0.4 MI SOUTH OF CR 357 TO 0.4 MI NORTH OF CR 357

5.1.4 Begin and End Callouts

The begin and end callouts for the project must include specific information. This should be formatted as follows:

- **Begin or End Project:** Identify whether the callout refers to the start or end of the project.
- **CCCC-SS-JJJ (CSJ Numbering System):** This code refers to the County Code (CCCC), State Segment Number (SS), and Job Number (JJJ). If the project has multiple CSJs, this should be shown for each CSJ involved.
- **STA (Station Number):** The station number indicating the exact location along the project length, with the stationing given in station+distance format.
- **REF MRK (Reference Marker):** The coordinates for the project's begin and end points should be included in decimal degrees format for precise geographic location tracking via Statewide Planning Map.

Note that station limits do not always correspond to the direction of the reference markers, regarding of whether the stationing is increasing or decreasing. The reference markers define the project's reference points, and this ensures accuracy when locating project limits.



5.1.5 Calculating Project Length with Bridges

When calculating the total project length, it is important to note that bridges should only be broken out separately if bridge-related work is included in the plans. For general purposes, the total project length is calculated from the begin to the end of the project, including both the roadway and any bridge structures.

WHARTON COUNTY

YOAKUM DISTRICT

EXCEPTIONS: NONE

EQUATIONS: NONE

RAILROAD CROSSINGS: NONE AT GRADE (ONE PARALLEL WITHIN PROJECT LIMITS.)

By adhering to these formatting and detail requirements, the Title Sheet will provide a comprehensive and accurate representation of the project, serving as an essential reference for design, construction, and future project management.

<i>PROJECT LENGTH</i>			
<i>ROADWAY</i>	<i>=</i>	<i>4,280.00 FT</i>	<i>= 0.810 MI</i>
<i>BRIDGES</i>	<i>=</i>	<i>00.00 FT</i>	<i>= 0.000 MI</i>
<i>TOTAL</i>	<i>=</i>	<i>4,280.00 FT</i>	<i>= 0.810 MI</i>

5.2 Index of Sheet

The Index of Sheets is a critical component of the project plans and must be prepared using the automated Index of Sheets Excel spreadsheet, which is provided by the Design Division. This can be found [here](#). This spreadsheet ensures uniformity and consistency in the organization and presentation of plan sheets. Instructions for using the spreadsheet can be found on the Design Division webpage, along with links to the necessary tools. The spreadsheet should be used in conjunction with the pen table to properly number and print the plan sheets.

When preparing the Index of Sheets, it is essential that the plan sheet names listed in the index exactly match those shown in the title block of each individual sheet. Consistency is crucial to avoid confusion and ensure that the plan sheets are properly referenced.

In naming the sheets for the Index of Sheets, the titles should be general and concise. It is advisable to combine sheets where possible to minimize the length of the index. This helps streamline the document and makes it easier for reviewers to navigate.

If additional information is required in the title block (for example, sheet details or clarifications), this should be presented in smaller text and enclosed in parentheses. Information within parentheses is not required to be included in the Index of Sheets. Only the primary sheet title, as shown in the title block, should be listed in the index. This approach reduces unnecessary clutter and maintains clarity in the document.

By following these guidelines, the Index of Sheets will provide an organized, consistent reference to all project plan sheets, helping streamline both the design and review processes.

5.3 Typical Section

1. Side Slope of 6:1 is most preferable for Yoakum District; however, it is necessary to confirm for every project with the design manager and/or TP&D. The typical maximum side slope shall be 4:1, with the exception that 3:1 slopes may be permitted for back slopes when necessary to avoid acquiring additional right-of-way.
2. Density Control: Density control methods shall be applied to the placement of all roadway bases and embankments, unless treated otherwise. This does not apply to short or narrow placements.
3. Flex Base Payment:
 - a. For all new flex base, payment will be made per cubic yard (CY).
 - b. For projects involving a mixture of new and existing base, payment will be made per cubic yard (CY).
4. Prime Coats: A covered prime coat should be applied when the roadway will be subject to traffic. If the roadway will not be under traffic, a standard prime coat should be applied.
5. Cross Slopes for Off-System Bridge Projects:
 - Off-system bridge projects shall have a cross slope of 2.0%.
 - For concrete pavement projects, the lane cross slope should be maintained through the shoulder.
 - Surface slopes for thin overlays should match the existing surface.

- All slope specifications must be reviewed and approved during the project kick-off or Pre-Design meeting.
6. Asphalt Binder for Overlays and Mill and Inlays:
 - For overlays and mill & inlay projects on US 59, use PG76-22 for the surface course and PG64-22 for all level-up and base courses.
 - For higher volume roads (typically 2000 ADT or more) on US and State highways, PG70-22 should be used for the surface course, unless special conditions warrant PG76-22. PG64-22 should be used for level-up and base courses.
 - PG76-22 may be used for other roadways with heavy truck traffic or high-volume turning movements if approved.
 7. Surface Course for Overlays and Mill & Inlays: Two inches or less of Type D surface shall be used for all overlay and mill & inlay projects.
 8. Lane Width for Rehabilitation Projects: For all FM or proposed State highway rehabilitation projects (2R & 3R), provide 12-foot lanes unless full-width shoulders are present, in which case 12-foot lanes should be used.
 9. Pavement Width: For all 2R, 3R, and 4R projects, a minimum of 26 feet of pavement should be provided. For systemic projects, the pavement must be widened to a minimum of 28 feet.
 10. Ride Quality for Overlays: Use Surface Test Type B, Schedule 3, for ride quality on overlay projects. For hot mix overlays with a single surface, use Schedule 2. For projects involving two different surfaces, such as B&D, B&B, or D&D, apply Schedule 1.
 11. Sealing New Flex Base or Cement-Treated Material: All new flex base or cement-treated material (including rehabilitation) must be sealed with a prime or covered prime coat.
 12. Cement-Treated Flex Base Specifications:
 - Cement-treated flex base typically contains 3% cement.
 - The estimated unit weight of the flex base is 135 pounds per cubic foot (lbs/cf).
 13. Sub-Grade Treatment ($PI \geq 18$): Sub-grades with a Plasticity Index (PI) of 18 or greater should generally be treated with lime at a rate of 5%.
 14. Sub-Grade Treatment ($PI < 18$): Sub-grades with a PI of less than 18 should generally be treated with cement at a rate of 3%.

15. Sub-Base Treatment: Sub-bases should generally be treated with cement at a rate of 3%. For pavement structures without flex base, use 4% cement. The estimated unit weight of the sub-grade is typically 115 lbs/cf.
16. Asphalt Concrete Pavement (ACP) Rates: For ACP applications, the following rates should be used:
- For all depths: 110 pounds per square yard per inch (lbs/SY/IN).
17. Under-Seal Application: An under-seal should be applied to Type B hot mix courses and other projects as determined by the project staff, such as on milled surfaces.
18. Flexible Base for Off-System Bridge Projects: Off-system bridge projects should utilize Flexible Base Type D, Grade 1-2, or a type preferred by the Area Engineer.
19. Aggregate Options for AC-20XP and CRS-2P: ASPH (AC20-5TR OR AC-20XP OR CRS-2P)
20. Aggregate Options for RC-250: When using RC-250, only Type E or Type L aggregates should be offered.
21. Bridge Structure Specifications: For projects with a bridge structure, both the existing and proposed bridge typical sections must be provided. The existing typical section should include the existing bridge rail type and the bridge's load rating. The proposed typical section should indicate any surfacing changes across the bridge, whether any new or retrofitted bridge rails will be installed, and the bridge's structural capacity. If the existing bridge is receiving additional surfacing, a load analysis from the Bridge Division must be submitted.

5.4 Estimate & Quantity Sheet

5.4.1 Estimate Rates for Surfacing Courses

ITEM 247 FL BS (CMP IN PLC) (TY E GR1-2) (FINAL POS) 8”: This item refers to the placement of flexible base (FL BS) in place, using Type E, Grade 1-2 material. The thickness specified for the final position is generally 8 inches. The unit of measurement for this item is in cubic yards (cy), which quantifies the volume of material used in the construction process.

Example: If a roadway section is 10 feet wide and 100 feet long, the volume of flexible base required can be calculated as follows:

$$\begin{aligned}\text{Volume} &= \text{Length} \times \text{Width} \times \text{Thickness} \\ &= 100 \text{ ft} \times 10 \text{ ft} \times 812 \text{ ft} = 66.67 \text{ cubic yards (CY)}\end{aligned}$$

ITEM 275, CEMENT 135#/CF 3% (Cement Treat (Exist MTL) (6")): This item pertains to the cement treatment of existing materials, where the cement content is 135 pounds per cubic foot (135#/cf), with a cement concentration of 3%. The treatment thickness is specified as 6 inches. The unit of measurement for this item is in square yards (SY), which is a measure of surface area, indicating the amount of treatment applied per square yard.

Example: If an area of 500 square yards is to be treated, the total cement quantity required can be estimated by first calculating the volume of cement required for the treatment:

$$\text{Volume of Cement} = 500 \text{ SY} \times 6 \text{ inches} \times \left(\frac{1 \text{ ft}}{12 \text{ inches}}\right) = 250 \text{ cubic feet (CF)}$$

$$\text{Weight of Cement} = 250 \text{ CF} \times 135 \text{ pounds per CF} = 33,750 \text{ pounds of cement.}$$

ITEM 316, PRIME, ASPH (RC-250) 0.20 GAL/SY, AGGR (TY-E GR-5 SAC-B) ICY/140SY: This item refers to the application of prime coat asphalt (RC-250), which is applied at a rate of 0.20 gallons per square yard (gal/SY). The prime coat is applied over Type E, Grade 5 SAC-B aggregate at a rate of 1 cubic yard (CY) per 140 square yards (SY). The unit of measurement for the asphalt prime coat is in gallons (gal), while the aggregate is measured in cubic yards (CY).

ITEM 316, OSCT, ASPH (AC-20-5TR or AC-20XP or CRS-2P) 0.40 GAL/SY, AGGR (TY-PE GR-3 SAC-B) 1 CY/85 SY: This item involves the application of surface treatment asphalt (options include AC-20-5TR, AC-20XP, or CRS-2P) at a rate of 0.40 gallons per square yard (gal/SY). The surface treatment is applied over Type PE, Grade 3 SAC-B aggregate, at a rate of 1 cubic yard (CY) per 85 square yards (SY). The unit of measurement for the asphalt is in gallons (gal), while the aggregate is measured in cubic yards (CY).

ITEM 316, SEAL, ASPH (AC-20-5TR or AC-20XP or CRS-2P) 0.34 GAL/SY, AGGR (TY-PE GR-4 SAC-B) 1 CY/130 SY: This item refers to the application of an asphalt seal coat (options include AC-20-5TR, AC-20XP, or CRS-2P) applied at a rate of 0.34 gallons per square yard (gal/SY). The seal coat is applied over Type PE, Grade 4 SAC-B aggregate, at a rate of 1 cubic yard (CY) per 130 square yards (SY). The unit of measurement for the

asphalt is in gallons (gal), while the aggregate is measured in cubic yards (CY).

ITEM 354, PLANE ASPH CONC PAV (2"): This item refers to the milling or planning of asphalt concrete pavement to a specified thickness of 2 inches. The unit of measurement for this process is in square yards (SY), as it represents the surface area of pavement to be milled.

ITEM 3076, D-GR HMA TY-B SAC-B PG64-22 8": This item involves the placement of dense-graded hot mix asphalt (HMA), Type B SAC-B, with a binder grade of PG64-22 at a thickness of 8 inches. The unit of measurement for this asphalt placement is in tons (ton), as the material is measured by weight.

ITEM 3076, D-GR HMA TY-D SAC-B PG70-22 2": This item refers to the placement of dense-graded hot mix asphalt (HMA), Type D SAC-B, with a binder grade of PG70-22 at a thickness of 2 inches. The unit of measurement for this asphalt placement is also in tons (ton), indicating the weight of the material required.

PFC

1 3/4" will be a 168 LBS/SY rate with 6% asphalt and 94% aggregate combined.

5.4.2 Estimate & Quantity Sheet

After the estimate is entered into TxDOTCONNECT, the sheet can be printed directly to PDF. The PDF can then be inserted into the project plans.

Vegetative Watering (Item 168) for seeding and/or block sod areas should be estimated at a rate of 13.6 million gallons per acre (TGL/AC) per cycle, with a total of 3 watering cycles. This calculation ensures that the vegetation receives sufficient water for establishment.

To clarify, the total estimated watering requirement is as follows:

- *Watering rate per cycle: 13.6 TGL/AC*
- *Total watering requirement for 3 cycles:*
13.6 MG/AC x 3 cycles = 40.8 million gallons per acre (TGL/AC)

This estimation applies to all projects involving vegetative establishment through seeding or block sod, and it ensures that water requirements are met to support healthy

growth and establishment.

Environmental (SWP3) summaries shall have the following notes placed under/with it:

“LOCATIONS AND TYPES OF BMPS MAY REQUIRE ADJUSTMENTS PRIOR TO OR AFTER PLACEMENT AS DIRECTED BY THE ENGINEER. ADJUSTMENTS SHOULD BE MADE TO ENSURE BMPS ARE WORKING EFFECTIVELY AND MAINTAIN COMPLIANCE WITH THE CONSTRUCTION GENERAL PERMIT. NOTIFY THE ENGINEER PRIOR TO MAKING ADJUSTMENTS.”

A Comparative study must be needed between the quantity summaries with the plan layouts and typical sections to verify that all bid items are accounted for, and none are missing. This ensures consistency between the bid documentation and the project plans.

5.5 Metal Beam Guard Fence and Existing Bridge Rail

Length of Need Calculations: The directory Project Design Guide contains resources to assist with Length of Need calculations, determining the length of MBGF when attached to T631 rail, identifying types of MBGF and end treatments, upgrading bridge rail, and a crash wall presentation.

Bridge Class Structures Without Safety End Treatments: For bridge class structures that do not have safety end treatments and cannot be extended with such treatments, MBGF must be installed, ensuring a minimum clear roadway width of 28 feet.

Non-Compliant Bridge Rails: The basic rule for non-compliant bridge rails is: If 10% of one side of a bridge railing is damaged, or if 75 feet of the railing is damaged (whichever is less), then rail replacement is recommended.

T631 Rail: When adding T631 rail to an existing bridge or to the headwall of an existing box culvert, Item 451 Retrofit Rail (TY T631) must be used.

T631LS Rail: When using T631LS rail on a bridge, ensure at least 12.5 feet of MBGF is installed before the DAT (Designated Anchor Terminal) to meet the requirement for 2 CRT posts for anchoring the T631LS rail.

SGT Placement on Non-Divided Highways: For MBGF installations on non-divided highways, SGTs (Guardrails) must be placed on the downstream end of the rail, regardless of the clear zone.

Attachment of Thrie-Beam and SGT: Technically, the thrie-beam and SGT can be attached to each other; however, this must be evaluated on a case-by-case basis for each project. If any part of the SGT is required for the length of need, it is recommended to include a length of MBGF between the thrie-beam and the SGT. Downstream ends where minimal or no length of need is required are the most likely locations for mounting the thrie-beam and SGT adjacent to each other.

SGT Selections: SGT (12S)31-18, and SGT (15)31-20 should be included in the plans where SGTs are required.

5.6 Culvert Design & Culvert Layouts

Existing 15” RCP driveway pipes may remain in place if they are in good condition and are approved during the design review process. For all new culvert structures, the minimum diameter must be 18 inches, and smaller downstream structures should be replaced as necessary. If funds are available, existing metal pipes should be replaced. During design reviews involving roadway culvert layouts, it is essential to have the required culvert photos available for viewing, as specified in the Survey Needs Task Deliverable.

A culvert layout must be provided for each roadway cross culvert in the project, regardless of whether work is being performed on that specific culvert. These layouts should be drawn with equal horizontal and vertical scales and placed on proportional grids, arranged in station order from top to bottom of the sheet. For projects with roadway culvert work, a series of drawings showing the plan & profile view must be included. If Roadway Plan/Profile Sheets are not part of the plan set, Plan & Profile Layout Sheets should be included, showing the location of each roadway culvert along with labels as needed. Additional roadway features, such as MBGF layouts, may also be shown on these sheets.

Finally, on the Culvert Summary Sheet (BCS), use conventional rounding techniques for measurements, rounding up or down as necessary, to ensure consistency and clarity in the documentation.

5.7 Hydraulics on Existing Crossroad Structures

As outlined in the Hydraulic Manual, Chapter 3, Section 1: Overview, a hydraulic analysis is required for any culvert that is being replaced. For culvert extensions, a hydraulic analysis must be conducted if one or more of the following conditions apply:

- There are high water reports at the culvert location.
- The roadway profile is being altered.
- There are any maintenance issues or concerns.
- Engineering judgment determines that a hydraulic analysis is necessary for a specific site, whereas the culvert is being extend a total of 8'+

If some crossroad culverts are included in the plans without a hydraulic analysis, a note must be added under the culvert summary to explain why the analysis was not conducted. The note should be edited to include the appropriate year frequency based on the as-built plans. Additionally, the culvert summary sheet must be signed and sealed by the responsible engineer to confirm the review and approval of the details.

As stated in Chapter 3, Section 2 of the Hydraulic Manual, the complete replacement of an existing structure that has not shown a history of problems must be justified by a compelling reason. Simply stating that the structure does not meet current hydraulic criteria for new design is not sufficient justification for replacement.

5.8 Bridge Hydraulics

Bridge Rail Considerations: Bridge hydraulics should always account for a 100% solid bridge rail, regardless of the type of rail specified for placement on the bridge.

Preliminary Hydraulic Report and Scour Analysis: A preliminary hydraulic report with scour analysis must be prepared and submitted for district review along with the 30% submittal. An updated version of this report must also be submitted with the bridge layout documentation to the Bridge Division for review and approval prior to final bridge design.

PSN Numbers for New Projects: PSN numbers should be requested early in the project development process, as they are required to be populated in TxDOTCONNECT before submitting the Preliminary Bridge Layout Review (PBLR) to the Bridge Division.

Required Information on Bridge Layout Sheets: The following information must be shown on all bridge layout sheets:

- Design Speed = XX MPH
- ADT (Current Year) = XXXX
- ADT (Design Year) = XXXX
- Functional Classification = XXXXXXXXXXXXXXXX
- Existing NBI: 11-XXX-X-XXXX-XX-XXX
- Proposed NBI: 11-XXX-X-XXXX-XX-XXX

Bridge Quantity Tables and Grouping: Bridge estimated quantity tables should be grouped with all other quantity summaries in the general section of the plan set. The bearing seat elevation sheet should be grouped with the bridge section of the plan set.

Existing Bridges to Be Removed: Any existing bridges that are to be removed must be shown, labeled and length noted in both the plan view on the P&P sheet and the bridge layout sheet.

Phased Cross Sections for Bridge Projects: All bridge replacement and bridge widening projects should include phased cross sections. This should be discussed during project scoping and included in the Work Authorization if necessary.

Preliminary Bridge Layout Review (PBLR) and approval is required for all bridges and bridge class culverts, regardless of funding type and work type, before any structural detail work is performed. Submit bridge layouts along with Hydraulic & Hydrology calculations and the scour analyses for all bridges and bridge class culverts, approved by the District Bridge Engineer, to the Plan Review Branch of the Bridge Division Bridge Management Section. Email the submittal of the PBLR folder link in ProjectWise to BRG_PD_PSE@txdot.gov. Refer to the *PBLR and PS&E Review Standard Operating Procedures* for in depth information on the process at http://crossroads.org/brg/PD/documents/PBLRPSE_BRG_SOP_02_21_19.pdf. The SOP outlines the process, level of review, and guidelines to estimate the time frame that the review will take. These reviews are vital to the overall success and longevity of bridge and culvert structures as PBLR reviews focus on FHWA and TxDOT policy and practice adherence.

Preliminary Bridge Layout Reviews (PBLR): It is essential that PBLR approval is obtained prior to preparing any detailed bridge design. Additionally, the April 8, 2019 Bridge Division memo from Graham Bettis, PE outlines the requirements for Preliminary Bridge Layout Reviews (PBLR).

District Preferences for On-System Bridges: The district preference for on-system bridges is to design them to meet a minimum 25-year frequency. Detour structures must meet HS20 loading and be hydraulically efficient.

Bridge Approach Work Justification: For bridge approach work, refer to the Bridge Project Development Manual, Chapter 2, Section 3 for justifications. All justifications should be sent to the District Bridge Liaison, by the District Bridge Engineer.

Drilled Shaft Lengths and Borings: If bore holes cannot be completed prior to letting, the following note should be included on the bridge layout: Drilled shaft lengths shown are approximate for bidding purposes. Lengths may change based on borings acquired after project letting. The contractor shall contact the engineer to verify all drilled shaft lengths and tip elevations at least two weeks prior to drilling operations.

5.9 Striping, Delineators, and Signing

5.9.1 Striping

- Seal Coat surfaces in the Yoakum District should be striped with 100 mil striping, while Hot Mix Asphalt (HMA) surfaces should have 90 mil striping.
- All roadways that are 20 feet wide or greater must include edge striping.
- For projects with pavement width of 28 ft or greater, profile pavement markings should be used, unless the project is within city limits or has a posted speed of 45 mph or less. (Reference: PM (1)-22)
- Obliterating striping must be done using seal coat unless other methods are approved by the Yoakum District Traffic Section.
- For arrows, words, symbols, stop bars, crosswalks, school crossings, and railroad crossings, use prefabricated pavement markings (Item 668, TY C).
- Raised pavement markers are to be used as guidance only.
- Rumble strips and profile markings should not be used on roadways with speeds of 45 mph or less, nor should they be placed within city limits.
- This design guide focuses on the striping configurations for 4-lane undivided highways using TxDOT's PM(2)-22 (Pavement Markings) and FPM(2)-22 (Freeway Pavement Marking) guidelines. These standards address the marking patterns, lane configurations, and safety features

specific to undivided highways with two lanes in each direction, typically used in the Yoakum District.

5.10 Delineators and Object Markers

5.10.1 Mailbox Installation

- Single Mailbox: 0560 7004 MAILBOX INSTALL-S (WC-POST) TY 3
- Double Mailbox: 0560 7005 MAILBOX INSTALL-D (WC-POST) TY 3
- Multiple Mailbox: 0560 7001 MAILBOX INSTALL-M (TWG-POST) TY 1

5.10.2.2 Delineators

- For cross culverts at headwalls, use 0658-7061 INSTL OM ASSM (OM-2Y) (WC) GND.
- For MBGF on undivided highways, use 0658-7019 INSTL DEL ASSM (D-SW) SZ 1(BRF) GF2 (BI).
- For MBGF on divided highways:
 - ✓ Yellow: 0658-7036 INSTL DEL ASSM (D-SY) SZ 1(BRF) GF2
 - ✓ White: 0658-7018 INSTL DEL ASSM (D-SW) SZ 1(BRF) GF2
- For Concrete Rail:
 - ✓ 0658-7013 INSTL DEL ASSM (D-SW) SZ 1(BRF) CTB (BI) for general concrete rail.
 - ✓ 0658-7031 INSTL DEL ASSM (D-SY) SZ 1(BRF) CTB for divided highways (yellow).
 - ✓ 0658-7012 INSTL DEL ASSM (D-SW) SZ 1(BRF) CTB for divided highways (white).
- For Off-System Bridge Rail and MBGF, use:
 - ✓ MBGF: 0658-7019 INSTL DEL ASSM (D-SW) SZ 1(BRF) GF2 (BI)
 - ✓ For DAT end, use 0658-7066 INSTL OM ASSM (OM-3L) (TWT) GND.
 - ✓ For Concrete Bridge Rail, use 0658-7012 INSTL DEL ASSM (D-SW) SZ 1(BRF) CTB.

5.10.3 Signing

- Ensure that the size of signing required is verified based on the type of roadway and district preferences in the Yoakum District.

- Stop signs must be a minimum of 48" x 48", unless special circumstances require a larger size. All stop signs must have reflective tape, and a general note on the plans should reference this requirement (from the district's master general note file).
 - Warning signs for on-system roadways must be at least 48" x 48". For expressways on US 59, use expressway-size signs.
 - For roadway culverts with headwalls that are 20 feet or longer, an object marker should be installed at each end of the headwall/wingwall.
- ❖ The size of signs depends on the type of roadway. For specific requirements, refer to the appropriate sign manual.

5.11 SW3P

5.11.1 Storm Water Pollution Prevention Plan (SWP3) Summary Sheet

The SWP3 Summary (Index) sheet is a required element for all PS&E (Plans, Specifications, and Estimates) submittals. The appropriate SWP3 summary sheet can be downloaded from the Roadway Standards website.

- 2023 SWP3 Summary Sheets (for projects less than 1 acre in size)
 - 2023 SWP3 Summary Sheets (for projects 1 acre or greater in size)
- *** (Anything over 10 ACRES disturbed will need to be sign and sealed)

Selection of the correct summary sheet is based on the total project area, in accordance with stormwater management requirements.

5.11.2 Soil Retention and Erosion Control

All slopes with a ratio of 4:1 or steeper and a height of 6 feet or more must be protected using Soil Retention Blankets. This measure is necessary to prevent erosion and ensure proper stabilization of the soil on the project site.

5.11.3 Environmental Layouts

- If the disturbance area for a project exceeds the threshold and requires an Environmental Layout Sheet, it must be included in the plans and properly labeled as such.

- Environmental Layout Sheets are required when significant ground disturbance will occur, such as for large-scale grading or drainage work, and are named accordingly in the plans. These sheets provide detailed information about erosion control measures and stormwater management.
- Plan Layout Sheets may be required if the disturbance volumes do not meet the threshold for SWP3 Layouts but still involve minor crossroad work or MBGF (Median Barrier Guard Fence) installation. These layouts, while related to environmental concerns, are typically less extensive and provide a basic layout of the proposed changes or installations.

5.11.4 EPIC Sheets

EPIC sheets (Environmental Protection Information and Compliance Sheets) are a mandatory part of the plan set for every project and will be prepared by the district environmental section. These sheets outline the environmental protections in place, including measures for erosion control, stormwater management, and any other environmental considerations relevant to the project.

5.11.5 Stream Names and Segments

For assistance with identifying stream names and segments relevant to the project, use the following TCEQ (Texas Commission on Environmental Quality) online map: [TCEQ Stream Segments and Names](#). This website allows the user to search for specific streams and their associated segments in the Yoakum District to ensure proper environmental and water quality compliance.

Key Notes: The Yoakum District adheres to state standards for stormwater pollution prevention and erosion control. Ensuring that all required SWP3 documentation, Environmental Layout Sheets, and EPIC Sheets are properly included in the plan set will ensure the project meets environmental regulations and compliance requirements. If Environmental Layout Sheets are not required but there is still a need to show smaller-scale impacts like crossroad or guardrail work, they will be labeled as Plan Layouts.

5.12 Truck Mounted Attenuators

5.12.1 Payment and General Notes

- TMA's (Truck Mounted Attenuators) are to be paid for under Item 505. Refer to the Yoakum District Master General Notes for specific general notes that must be included in the PS&E (Plans, Specifications, and Estimates) for the project.
- Payment for TMA's will be made on a per day basis. This applies to the setup, operation, and monitoring of traffic control zones throughout the duration of the project.

5.12.2 Seal Coat and Moving Operations

- Use a Mobile TMA for tasks such as RPM removal, striping, and other moving operations. This will be applicable to striping-related work and any other tasks where traffic control zones need to be mobile and adjusted as the work progresses along the roadway.

5.12.3 Additional Resources

For further guidance on TMA's and their application on the project, consult the PS&E Preparation Manual, specifically Section 4.5.5, which provides more detailed instructions on their proper use and management.

5.13 Cross Sections

5.13.1 General Cross Section Requirements

- Cross sections must be drawn at actual size in the master file. While scaling can be done when placing them on the sheets, the district preference is to use the same scale both horizontally and vertically. This helps avoid exaggerated vertical scales that could distort the true representation of the roadway.
- When preparing the cross sections, it's important to verify that the cut and fill lines/areas are modeled accurately for each layer of the pavement structure, including the selection of fill

material, unsuitable material, and other components. Proper modeling ensures accurate calculations for earthwork quantities.

5.13.2 Cross Section vs. Master Plan

- Always compare the cross sections with the master plan drawing to ensure that the road widths, transitions, superelevations (supers), and other features match correctly between the two documents. Any discrepancies should be addressed before finalizing the plans.

5.13.3 Final Plan Set Requirements

Cross sections will not be included in the final plan set; however, the following requirements must be met:

- Title Block and Sheet Title: Each cross-section plan sheet must include a full title block and a sheet title.
- Labeling for Contractor's Information: Each cross-section plan sheet must be labeled "For Contractor's Information Only." This clarifies that the sheets are intended to provide additional details for construction purposes.
- Bridge Replacement Projects: For projects involving bridge replacements, TCP/Phased cross sections are required. These will reflect the various stages of construction, including any necessary lane closures, detours, and staging.

By following these guidelines, the cross sections will provide an accurate and useful representation of the project's design, supporting both construction and quality assurance efforts.

Construction

6.1 Milestones, Substantial Completion and Project Liquidated Damages

6.1.1 Complete Form 2699

Form 2699 identifies situations where an Incentive or Additional Project-Specific Liquidated Damages (APSLD) may be applicable.

6.2.4 Estimate

- Include any incentives in the Engineer's Estimate.
- Incentives are not bid items (such as force accounts) but should be accounted for as part of the project cost.
- These costs will be charged to the district budget during construction.

6.2.5 General Notes

- The Master General Notes include templates for milestones and substantial completion, as referenced in Item 8.

6.2.6 RUC (Road User Cost) Calculation

- Construction Division RUC Calculator: The Construction Division has an online RUC calculator available at: [TxDOT RUC Calculator](#).

6.2.7 Maximum Incentive Amount

- Total Incentives: The total amount of all incentives is recommended not to exceed 3.4% of the total construction estimate. This is based on an evaluation of Missouri DOT projects by TTI (Texas A&M Transportation Institute).
- FHWA Guidelines: The Federal Highway Administration (FHWA) recommends that the total amount for incentives should not exceed 5% of the total contract value.

6.2.8 Off-System Bridges and Construction Scheduling

When scheduling the construction of Off-System Bridges, it's important to consider the following:

- Higher Liquidated Damages (LDs): Off-System Bridges carry higher LDs compared to typical projects. This means penalties for delays are significantly more severe.

- **Impact on Project Timelines:** Any delay in completing the bridge work can result in substantial financial penalties. Careful planning is essential to avoid these costs.
- **Critical Scheduling Considerations:** Ensure that the schedule accounts for factors such as:
 - Weather conditions
 - Resource availability
 - Site-specific challenges
- **Close Monitoring:** Regular updates and monitoring are crucial to keeping the project on track and preventing delays that could lead to higher LDs.
- **Manage the Critical Path:** Prioritize activities that are on the critical path to ensure timely completion and avoid any schedule overruns.

By factoring in these considerations, potential risks can be mitigated, and the likelihood of incurring additional costs due to delays can be reduced.

6.3 CPM or Bar Chart

Standard Specification Item 8.5.2 allows the use of either a Critical Path Method (CPM) or a Bar Chart for scheduling. However, there are specific cases where a CPM schedule is required in the Master General Notes for certain projects.

6.3.1 Master General Notes – CPM Schedule Requirement

- The Master General Notes should specify the requirement for a CPM schedule under the following conditions:
 - ❖ The project is estimated to be over \$25 million.
 - ❖ The project is under \$25 million but includes any of the following elements:
 1. Substantial Completion Milestone.
 2. A+B Bidding Provisions (Time and Cost Incentive/Disincentive).
 3. Conditions that indicate a high probability of project delay (such as complex or time-sensitive elements).
 4. The project includes a management plan for utilities.
 5. The project has ROW (Right-of-Way) parcels that are not acquired by the time of letting.
 6. If specifically requested by the Area Engineer (AE).

6.3.2. Deletion of CPM Schedule Requirement

- The Master General Notes should not require a CPM schedule under the following circumstances:
 1. The project is under \$25 million and does not include any of the above-listed elements.
 2. The project is an overlay or seal coat.
 3. The project is an SET (Safety) project.

Summary: CPM Schedule is required for large or complex projects with high potential for delays or specialized requirements. For projects under \$25 million, the CPM schedule requirement is based on specific conditions (e.g., substantial milestones, A+B bidding, utility management plans, etc.). For smaller, simpler projects, such as overlays, seal coats, and SET safety projects, the CPM schedule requirement can be deleted.

6.4 A+B Bidding

The A+B bidding method is a valuable tool for time-critical projects or projects with critical milestones. Under this approach, the contractor's bid is evaluated based on both the cost of the work (A) and the time required to complete the project (B). This method encourages contractors to minimize construction time while maintaining cost effectiveness.

It is important that A+B bidding is discussed throughout the project development process, from early planning to final design. Key considerations should include:

- Identifying critical milestones and deadlines.
- Evaluating the impact of construction time on overall project success.
- Collaborating with the contractor to ensure realistic timelines and achievable project goals.

By incorporating A+B bidding, project teams can better manage time-sensitive elements and ensure that both cost and time are optimized to meet project goals and milestones.

6.5.2. Utilities and Right-of-Way (ROW)

- Include specific activities related to each utility conflict or utility owner that impacts the project timeline.
 - Ensure all utility relocations and necessary ROW activities are properly accounted for within the schedule.

6.5.3. Cure Time

- Be sure to include appropriate cure times for:
 - Concrete structures – Account for curing durations as per project specifications.
 - Cement and lime-treated materials – Include necessary cure times for these materials to ensure structural integrity.

6.6 Computation of Contract Time

6.6.1. Standard Workweek – 8.3.1.4

All projects should follow the Standard Workweek of five days per week, unless otherwise approved.

6.6.3. Use of Six-Day Workweek

- A Six-Day Workweek may be used but requires DCO (District Construction Office) approval.
 - Six-day weeks should be justified with appropriate planning and scheduling considerations, including time for weather delays, holidays, and any potential unforeseen interruptions.

6.6.4. Prohibited Workweek Types

- Seven-Day Workweek and Calendar Day schedules are not allowed without specific, unusual circumstances and explicit approval from higher authorities.

6.7 Delayed Start Special Provisions

6.7.1. Asphalt Projects

- SP 008-002 must be included in all overlay and seal coat projects.
 - ✓ Purpose: This SP allows the contractor to select a start date for work during the asphalt season based on their preference.
 - ✓ Asphalt Season: The start and end dates for the asphalt season are specified in the Master General Notes under Item 300.
 - ✓ Latest Start Date: The latest work start date is provided in the Master General Notes under Item 8.

TxDOT Connect Information:

- Change Summary Section: No changes proposed for SP 008-002.
- Comments Section: SP 008-002 enables the placement of asphalt products to align with the dedicated asphalt season. The Yoakum District will specify the asphalt season and latest

work start date in Item 300 and Item 8 of the general notes for all overlay and seal coat projects.

6.7.2. Signal and Illumination Projects

- SP 008-005 should be included in all signal and illumination projects to allow a 90-day delay.
 - ✓ Purpose: This provision accounts for the lead times associated with metal and electrical components, which can significantly delay project progress.

TxDOT Connect Information:

- Change Summary Section: No changes are proposed to SP 008-005.
- Comments Section: SP 008-005 allows the contractor to order and take delivery of materials for signal and illumination work, including electrical meters from local utilities. These materials often have long lead times (up to 6 months). This provision ensures that electrical services can be made available once the electrical work is completed, addressing the industry-wide supply chain delays.

6.7.3. Joint Bid Utilities Projects

- SP 008-003 should be included in all projects with joint bid utility work, allowing for a 90-day convenience delay.
 - ✓ Purpose: This delay gives the contractor time to submit materials for approval by the local utility owner, ensuring that utility components are approved before purchase.
 - ✓ Industry Issue: Long lead times between purchase and delivery of utility materials make this delay necessary to ensure proper procurement and timely delivery of materials.

TxDOT Connect Information:

- Change Summary Section: No changes are proposed to SP 008-003.
- Comments Section: SP 008-003 permits contractors to prepare for joint bid utility work, allowing the delay period to be used for obtaining utility material approval, purchasing, and delivery in advance of construction.

6.7.4. 60-Day Delay

- Use of 60-Day Delay instead of SP 008-002:
 - ✓ If a 60-day delay is required for a project, SP 008-002 should be replaced with a 60-day delay provision for better flexibility.

6.7.5. 120-Day Delay

- Use 120-Day Delay instead of SP 008-004:
 - ✓ When a 120-day delay is needed, replace it with SP 008-004 for extended timelines.

6.7.6. 180-Day Delay

- Use SP 008-010 for projects requiring a 180-day delay. This is applicable for projects with significant delays needing extended time beyond the standard 120-day provisions.

Project-Specific Guides

7.1 Mobility Projects and Non-Overlay Projects

7.1.1 Mobility Projects

Mobility projects typically involve capacity expansion, intersection improvements, new lanes, bypass construction, and other improvements aimed at increasing traffic flow and reducing congestion.

Key Design Guidelines for Mobility Projects:

- Alignment and Geometry:
 - ✓ Ensure proper horizontal and vertical alignment to accommodate design speed, sight distance, and safe turning radii.
 - ✓ Superelevation design should be based on TxDOT standards, and transitions between superelevated sections should be gradual.
 - ✓ Consider cross-sectional elements such as lane widths, shoulder widths, and medians, particularly when planning for multi-lane roads or new interchanges.
- Roadway Sections:
 - ✓ For new lanes, reconstruction, or bypass roads, use appropriate roadway sections (e.g., standard rural or urban cross sections) depending on the project location.
 - ✓ Follow minimum lane widths per TxDOT standards and ensure adequate clear zones for roadside safety.
- Traffic Control Plans (TCP):
 - ✓ Develop detailed traffic control plans for all phases of construction to ensure safe and effective movement of vehicles and pedestrians during the project.
 - ✓ For work zones impacting high-traffic areas, ensure advance warning signs, temporary barriers, and flagger operations are carefully planned.
- Drainage:
 - ✓ Proper drainage design is essential, particularly when adding lanes or modifying road profiles. Ensure that roadway culverts, drainage inlets, and ditch profiles are sized appropriately.
 - ✓ Hydraulic analysis may be required depending on project scope and location.
- Mobility Enhancements:

- ✓ Incorporate smart technologies (e.g., signal timing optimization, dynamic message signs, automated traffic management systems) where feasible to increase mobility efficiency.
- ✓ Consider non-motorized users, including bicycle lanes and pedestrian facilities, where applicable.
- Interchanges: When designing new interchanges, consider:
 - ✓ Adequate acceleration and deceleration lanes.
 - ✓ Appropriate turning radii and flared ramps.
 - ✓ Incorporation of loop ramps and flyover bridges where necessary.
- Signalized Intersections:
 - ✓ When upgrading or installing new signalized intersections, ensure that signal phasing is designed to minimize delay and improve traffic flow.
 - ✓ Implement adaptive signal control technology (if applicable) to adjust signal timings based on real-time traffic conditions.
- Intelligent Transportation Systems (ITS):
 - For larger mobility projects, integrate ITS infrastructure (e.g., traffic cameras, vehicle detection systems, variable message signs) to improve traffic flow and safety.

7.1.2 Non-Overlay Projects

Non-overlay projects involve activities such as maintenance (e.g., seal coats, pothole repair), shoulder widening, reconstruction of existing roadways, or spot improvements. These projects typically do not involve substantial changes in the cross-section or capacity of the road.

Key Design Guidelines for Non-Overlay Projects:

- Surface Treatment and Pavement Design:
 - ✓ If the project includes seal coating, follow the appropriate specifications for materials, application methods, and timing (refer to SP 008-002 for asphalt season).
 - ✓ For pothole repairs or localized reconstruction, ensure the patching or replacement materials meet the specifications outlined in the TxDOT Roadway Design Manual.

- Shoulder Widening:
 - ✓ If the project includes widening of shoulders, ensure the shoulder design is consistent with TxDOT standards (i.e., 8-12 feet for rural highways, 5 feet minimum for urban roads).
 - ✓ Consider drainage impacts when widening shoulders and ensure that shoulder drainage systems are updated accordingly.
- Roadway Widening or Realignment:
 - ✓ When widening or realigning an existing roadway, ensure that roadway geometry (i.e., cross slope, superelevation, and vertical curves) aligns with existing or future traffic demands.
 - ✓ Design transition zones where the widening or realignment occurs to minimize abrupt changes in road geometry.
- Cross-Sectional Adjustments:
 - ✓ For non-overlay projects, if a cross-sectional change is required (e.g., adding a lane, creating a dedicated turn lane), ensure that all design elements (such as lane width, shoulder width, and median width) comply with TxDOT Roadway Standards.
 - ✓ Pay special attention to the clear zone and ensure adequate safety features like guardrails, barriers, or end treatments are incorporated.
- Geotechnical Considerations:
 - ✓ For projects involving substantial excavation or new embankments, a geotechnical investigation should be performed to determine the soil conditions and the need for stabilization (e.g., lime treatment, geosynthetics).
- Utilities and ROW:
 - ✓ Ensure all utility relocations are planned and coordinated with utility companies.
 - ✓ If right-of-way (ROW) acquisition is required, the schedule for acquisition must be considered, and all parcels should be in place before construction starts.
- Traffic Control and Safety:
 - ✓ For non-overlay projects, especially those involving short-term lane closures or restricted access, ensure the Traffic Control Plan (TCP) includes:

- ✓ Proper placement of signage, barricades, and flaggers to ensure traffic safety and minimize disruption, however, only state roads are to be included in signed detour routes.
- ✓ For projects with significant disruption, detour routes must be clearly signed and communicated to the public.
- **Special Considerations for Non-Overlay Projects:**
 - ✓ **Spot Improvements:** Small but critical enhancements like intersection improvements, spot repairs, or traffic signal installations. Consider both short-term and long-term impacts on mobility and safety.
 - ✓ **Bridge Repairs:** If the project includes bridge deck repairs or bridge widening, it is essential to consult and follow the District Bridge Engineer's specific guidance to off-system bridges in the Yoakum District.

7.1.3 General Notes and Requirements for Both Mobility and Non-Overlay Projects

- **Environmental Considerations:**
 - ✓ For both mobility and non-overlay projects, ensure compliance with SWP3 and Environmental Layouts (refer to Environmental Section guidance).
 - ✓ Projects near wetlands or sensitive habitats require environmental clearance before work can begin.
- **Funding and Budgeting:**
 - ✓ All project estimates must clearly include contingencies for unforeseen conditions (e.g., soil conditions, utility conflicts).
 - ✓ Ensure accurate budget forecasts for both construction and ROW acquisition.
- **Coordination with Local Agencies:**
 - ✓ Coordinate with local municipalities and emergency services for mobility projects, especially if the project affects heavily trafficked routes or key intersections.
- **Submittals and Reviews:**
 - ✓ Submit CPM schedules for projects exceeding 24 months and ensure DCO reviews for schedule and time extensions.
 - ✓ Geotechnical and environmental reviews should be completed early in the design phase to prevent delays during construction.

7.2 Overlay/Seal Coat Projects

7.2.1. Pavement Design and Preventive Maintenance

7.2.1.1 Reference the District Pavement Design SOP:

- Review the Pavement Design Approach outlined in the District Pavement Design SOP, particularly for preventive maintenance and rehabilitation projects. Ensure the approach is applied correctly for each type of project.
- Follow the District Material Selection Guide (Attachment 2 of the SOP) for selecting appropriate materials for pavement design and rehabilitation.

7.2.1.2 Project Site Visit:

- Prior to design, visit the project site to assess areas that need improvement (e.g., rough pavement, drainage issues, or visible signs of deterioration).
- Collaborate with the Area Office and Maintenance Section Supervisor to identify additional issues that may need correction, such as potholes, ruts, or drainage blockages.

7.2.1.1 Special Provisions and Delayed Start:

- Include the Delayed Start Special Provision for asphalt season as part of the project documentation. Reference the Delayed Start section in this guide.
- Ensure that the work start date is consistent with the asphalt season as per TxDOT standards.

7.2.3.2 Backfill Pavement Edges:

- Use Item 134 for Type A Backfill unless working with a curbed roadway.
- Include a note in the sequence of work to place backfill prior to surface application.

7.2.3.3 Pavement Repairs:

- Include Item 351 for pavement repair where there are cracks, ruts, or other significant pavement distress that need to be fixed before resurfacing.

7.2.3.4 Level-Up:

- Level-up with Type D material when needed to restore cross slopes and correct rutting.
- Include 10% of surface quantity for all roads showing visible rutting, flat cross slopes, or a bumpy ride (where overlay thickness is 1 inch or less).

7.2.3.5 Milling:

- Use Item 354 for milling operations.

- For milling under bridges/structures, ensure that milling maintains the vertical clearance as specified. Maintain a +3” buffer beyond the signed vertical clearance to avoid obstruction.

7.2.4. Asphalt and Pavement Preservation

7.3.4.1 Asphalt Projects:

- Include SP 008-002 for overlay and seal coat projects to allow work during the asphalt season.
- Asphalt season details are outlined in the Master General Notes under Item 300.

7.2.4.2 Bridge Decks:

- Bridge Deck Overlays: Contact the Yoakum District Bridge Section for coordination on bridge deck overlays.
- For cleaning and sealing of existing joints, include Item 438.
- Use Item 785 for the repair/replacement of existing joints.
- Do not overlay or use seal coat on direct traffic concrete bridge decks.

7.2.5. Traffic Control and Intersections

7.2.5.1 Traffic Signals:

- For signal upgrades or new installations, include Item 688 to replace existing signal loops or upgrade to an alternative detection method.

7.2.5.2 Metal Beam Guard Fence (MBGF):

- Update MBGF to meet the new standards and correct height specifications. Refer to project-specific guidelines for adjusting or updating MBGF.

7.2.5.3 Guardrail End Treatments:

- Ensure all guardrail end treatments are updated to the new standards with correct height as per TxDOT specifications.

7.2.5.4 Intersections and Driveways:

- For intersection or driveway paving, include Item 530-7022 and provide a plan detail showing tapers for proper integration into the roadway system.

7.2.6. Erosion Control and Floodplain Considerations

7.2.6.1 Erosion Control:

- Include Item 506 for erosion control as necessary based on site conditions, such as steep slopes, disturbed areas, or areas near water bodies.

7.2.6.2 Floodplain Considerations:

- Avoid raising the roadway profile in floodplain areas.
- If profile raising is necessary, coordinate with FEMA floodplain officials to ensure No Adverse Impact as required by federal regulations.
- Provide a mill detail to maintain the existing profile elevation or provide proper mitigation strategies.

7.2.7. Operational and Design Improvements

7.2.7.1 Operational Improvements:

- Consider modifications to striping to improve operational efficiency, particularly for high-traffic areas, intersections, or turn lanes.
- Modify striping for improved bicycle accommodations or to address safety concerns in areas with high pedestrian traffic.

7.2.7.2 Tree Trimming and ROW Issues:

- Consult the Area Office to determine if tree trimming is required, and whether Item 100 (ROW preparation) and “PREP ROW PRUNING DETAIL” are necessary for ROW clearing and preparation.

7.3 Off-System and Small Bridge Projects

The use of default beams, riprap, and bridge rails ensures standardization across projects, while the guidelines for cost reduction and embankment specifications offer ways to reduce expenses without compromising quality. Proper attention to traffic control, contract time schedules, and milestone management ensures that construction progresses smoothly and on time.

7.3.1. Bridge Beams

Ensure that the selected beam type is appropriate for the required span and structural load.

7.3.2. Bridge Abutment Riprap

To protect bridge abutments from erosion, it is essential to include Item 432, which covers the installation of riprap. Riprap helps stabilize the soil around the abutment and ensures long-term durability of the structure against water flow and other environmental factors.

7.3.3. Bridge Rail

For the installation of bridge rails, use Item 450. This item includes the necessary materials and procedures for ensuring that the bridge rail meets the safety and design standards required for protecting vehicles and pedestrians.

7.3.4. Pavement Section Guide

For a comprehensive guide on the proper selection of pavement sections of the project, refer to the Pavement Section Guide located in ProjectWise at:

13 - YKM\Offices\Construction\Designers Guide\Pavement Design\.

This guide provides detailed specifications and recommendations on pavement types based on the scope of the project and traffic considerations.

7.3.5. Cost Reduction Measures

To reduce project costs while maintaining the integrity of the road structure:

- Item 247 (Flex Base) should be placed using ordinary compaction techniques instead of more costly methods, thus lowering construction costs without compromising the quality of the subgrade.
- Item 275 (Cement Treated Base) can be mixed with cement using a motor grader instead of more expensive equipment. This method is both cost-effective and efficient for rural areas or less trafficked roads.
- Item 341 allows the use of Type D (Exempt) hot-mix asphalt in rural areas where the ADT (Average Daily Traffic) is below 10,000. For such areas, using a motor grader to place the hot-mix provides cost savings.

7.3.6. Embankment

Use Type C Embankment as the standard material for embankment construction. This type of embankment provides adequate support for the roadway structure and is the most commonly used for embankment fill, ensuring stability and long-term performance of the roadbed.

This material is carefully selected to balance strength, compaction, and drainage characteristics, making it the most common choice for embankment fills in roadway construction.

For retaining wall applications, a different material and construction method are used. Please refer to specific retaining wall design guidelines and/or Engineer-in-charge for appropriate materials and techniques.

7.3.8. River Access

For projects requiring construction access to rivers or bodies of water, additional Pay Items may be necessary in the contract to account for the costs associated with constructing access roads and work areas. This provision ensures that the construction site is safely and efficiently accessible, especially when working near water bodies.

7.3.9. Contract Time Determination Schedule

The construction schedule should be based on a bar chart template to establish project timelines and track milestones. This schedule should accurately reflect the estimated construction time and help manage the sequence of work activities for timely project completion.

7.3.10. Traffic Control

Before proceeding with any closures or detours, verify with local authorities to confirm that road closures and detours are allowed in the project area.

7.3.11. Milestones

If the project requires a road closure, consider including milestones in the project schedule to control the closure duration.

7.4 Roundabout Projects

Roundabouts are a modern and efficient alternative to traditional signalized or stop-controlled intersections. They help improve traffic flow, reduce crashes, and lower vehicle emissions. Designing a roundabout requires careful consideration of several factors, including traffic patterns, geometry, pedestrian safety, and environmental impact.

Roundabouts on highways serve as effective intersections that manage traffic flow, reduce congestion, and improve safety. They are particularly useful on highways where high traffic volumes and safety concerns can be addressed without the need for signalized intersections. A

roundabout can be a more efficient and safer alternative to traditional intersections, especially on rural highways with moderate to high traffic volumes.

TxDOT Roadway Design Manual, AASHTO's Green Book, and any relevant Local Municipality Design Standards should be followed.

7.5 Culvert Widening, Installation, and Replacement

For culvert and pipe end protection, SETs should be considered as the first option, with MBGF serving as an alternative, though it does come with a long-term maintenance commitment. Temporary shoring is necessary for the installation of all headwalls requiring excavation at the back, as well as for any joints impacted by traffic control phasing. Excavation for widenings often creates a vertical face that requires shoring to facilitate culvert extensions and headwall construction. In cases where trench excavation protection is needed, Item 402-7001 should be included, especially in areas parallel to culvert extensions.

When planning traffic control, it's important to assess the need for lane reductions if the culvert work impacts traffic flow. Considerations should include whether the ADT (Average Daily Traffic) is too high for a one-lane, two-way traffic control setup, or if the work can be completed during a daytime work shift and the travel lane restored. If the work is extensive, completing it over a weekend might be necessary. The relevant pay items and notes should be added to the plans to provide guidance on handling traffic disruptions.

For projects where fence replacement is necessary, if the culvert or outfall work is within 5 feet of the ROW line, include a fence replacement item based on WF(1)-10 and WF(2)-10 standards, with Item 552-7003 as the default for new fence installations. Note that the removal of fences is considered subsidiary to Item 552.

The Temporary Signal Option for TCP (Traffic Control Plan) is used during culvert widening, installation, or replacement projects to manage traffic flow safely through construction zones. When roadwork requires lane closures or shifts, temporary traffic signals are deployed to alternate traffic in one direction at a time, particularly when only a single lane is available. These signals help maintain smooth traffic flow, reduce congestion, and ensure safety for both drivers and workers. By controlling the timing and direction of traffic, temporary signals minimize delays, prevent accidents, and allow construction to proceed efficiently, all while keeping vehicles and pedestrians safe in the vicinity of the work zone.

7.6 Notch and Widening Projects

If the vertical and horizontal profile of the roadway does not require adjustments, refer to the General Plan Sheet Preparation Guide for guidance on the vertical profile and horizontal alignment, ensuring proper reference for alignment and grading.

For pavement repair, include Item 351 for all necessary pavement repairs, with quantity and depth specified by the Pavement or Construction Engineer.

7.10 Water/River/Creek/Stream Projects

For projects that involve river, creek, or stream crossings, specific provisions and guidelines must be followed to ensure proper handling of waterway access, environmental impact, and construction activities. Below is a summary of the necessary design steps and requirements for river access, temporary river impacts, cofferdams, and turbidity control measures for projects in the Yoakum District.

7.10.1. River Access and Temporary Construction Access

- **Temporary Construction Access :** For projects that require construction access to a river, creek, or stream, include - Temporary Construction Access in the plans. This item is used to accommodate temporary access points for construction equipment or staging in waterway areas.
- **Temporary River Impact or Access:** If the project involves any work within or near the river, notify DEQC (District Lead Environmental Planner) for coordination. The following scenarios require special attention:
 - ✓ **Bridge Work Within Ordinary High-Water Mark:** Any bridge work or construction activities that are within the limits of the ordinary high-water mark need to be addressed with DEQC and coordinated with the Environmental Division (ENV) for permits.
 - ✓ **Bridge Work in the Water:** If the bridge work involves activities in the river or stream, DEQC coordination is required.
 - ✓ **Disturbed Soil Near Ordinary High-Water Mark:** If construction activities disturb soil within 10 feet of the ordinary high-water mark, it will trigger additional environmental review and permitting.

7.10.2. Cofferdams

- Item 403-7002 - Temporary SPL Shoring (COFFERDAM): Use Item 403-7002 for the construction and installation of cofferdams to divert water flow during construction activities in or near the waterway.
- General Notes for Cofferdams: Include the following notes for the proper installation, design, and payment of cofferdams:
 - ✓ Contractor's Responsibility: The contractor is responsible for designing the temporary cofferdam and ensuring it is capable of handling the required height and water flow. The height indicated in the plans is an engineer's estimate for quantity purposes and should not be considered as the final design. If the cofferdam needs to be raised to accommodate changes in water levels, the contractor will not receive additional compensation beyond the original bid quantity.
 - ✓ Adjustment for Riverbed Elevation: The contractor must also account for variations in the riverbed elevation. The plan shows an estimated elevation for the riverbed; however, the actual elevation may vary. The contractor is responsible for adjusting the cofferdam design to accommodate the real riverbed elevation and will not receive additional compensation for these adjustments.

7.11 Pedestrians

For projects that involve work near existing pedestrian paths or sidewalks, special attention must be given to maintaining pedestrian safety and accessibility throughout the construction process. Below are the guidelines and required provisions for addressing pedestrian impacts, providing temporary pathways, and ensuring compliance with pedestrian-related requirements in the Yoakum District.

7.11.1. Impacts to Existing Pedestrian Paths

- Non-ADA Compliant Paths: Any disruption to existing pedestrian pathways that do not meet Americans with Disabilities Act (ADA) standards must be addressed. This includes temporary paths that may be created due to construction traffic, detours, or other construction-related activities.
- Alternate Route or Temporary Path: If the existing pedestrian route is interrupted or unavailable due to construction, a temporary, safe alternate route must be provided. This alternate path should be clearly marked, accessible, and designed to minimize inconvenience for pedestrians, particularly for individuals with disabilities.

7.11.2. General ADA Compliance Requirements

- **ADA Standards:** All curb ramps, sidewalks, and pedestrian pathways must comply with the 2010 ADA Standards for Accessible Design (accessible via the U.S. Access Board website). These standards include specific requirements for dimensions, slopes, and cross-slopes.
- **Pathway Continuity:** Pedestrian routes must be continuous and accessible from one point to another. Ensure that sidewalk paths and curb ramps are integrated into the design to maintain uninterrupted access for individuals using mobility aids, such as wheelchairs, walkers, and other assistive devices.

7.11.3. Curb Ramp Design

- **Location and Placement:** Curb ramps should be placed at all pedestrian crossings to provide direct access from the sidewalk to the street, in alignment with the pedestrian pathway. Ramps must not be obstructed by utility poles, trees, or other elements that can impede safe passage.
- **Ramp Dimensions:**
 - ✓ **Width:** A minimum clear width of 36 inches is required for all curb ramps.
- **Slope:** The running slope (the slope of the ramp) should be no steeper than 1:12, meaning for every inch of height, the ramp must extend 12 inches horizontally.
- **Cross-Slope:** The cross-slope of the ramp should be no steeper than 1:48. This ensures that the ramp does not tilt sideways, which could cause a safety hazard.
- **Landing Area:** The landing area at the top and bottom of the ramp should be level and at least 36 inches wide by 36 inches long. It should provide a safe and stable area for pedestrians to pause, turn, or maneuver in and out of the crosswalk.
- **Detectable Warnings:** Include truncated domes (detectable warnings) at the base of the curb ramp, flush with the surface, to alert visually impaired pedestrians of the upcoming street or roadway. The truncated domes must meet ADA requirements for size, spacing, and contrast.

7.11.4. Sidewalk Design

- **Clear Width:** The minimum clear width for sidewalks should be 5 feet, but in areas with high pedestrian traffic or in commercial zones, a wider width of 6 to 8 feet may be necessary to accommodate wheelchairs and other mobility aids.

- **Surface:** Sidewalks should have a smooth, firm, and stable surface. Avoid cracks, holes, or other defects that could impede mobility. Surfaces such as concrete, asphalt, or compacted stone are preferred, with texture applied only to ensure traction, not to create barriers.
- **Cross-Slope:** The sidewalk's cross-slope should not exceed 2% (1:50), which is the maximum allowable slope to ensure proper drainage while preventing accessibility issues.
- **Obstructions:** Ensure that sidewalks are free from obstructions such as utility poles, tree roots, signs, or other elements that could create barriers to pedestrians. If obstructions are unavoidable, clear zones of at least 36 inches should be maintained for wheelchair access.

7.11.5. Ramps at Intersections

- **Parallel Ramp Configuration:** Where possible, curb ramps should be configured parallel to the direction of travel to minimize slope and maximize stability. At T-intersections or crosswalks, ensure that ramps align with the pedestrian path to reduce turning movements.
- **Turning Radius:** At intersections, ensure that the turning radius at the bottom of the ramp is wide enough to allow individuals using mobility aids to turn easily. A minimum turning radius of 48 inches is recommended.
- **Crosswalk Alignment:** Ensure that ramps align directly with the crosswalk to avoid any detours that would force individuals to walk further to cross safely.

7.11.6 Inspections and Compliance

- **Field Inspections:** Ensure that curb ramps and sidewalks are inspected regularly during the design, construction, and post-construction phases to verify compliance with ADA standards. This includes confirming the slope, width, cross-slope, and surface conditions.
- **Coordination with Local Entities:** For large-scale projects or where significant changes are being made to existing infrastructure, work closely with local disability advocacy groups and local government agencies to ensure that pedestrian access needs are properly addressed.
- **Post-Construction Review:** After construction, conduct a final review to ensure all pedestrian paths and ramps are ADA-compliant. Any deviations from the design or issues with implementation should be corrected promptly.

7.12 Pavement Design for Off-System Bridges

The design of pavement for off-system bridges is a critical aspect of ensuring both the safety and longevity of the structure, as well as its ability to accommodate traffic loads and

environmental conditions. Off-system bridges, typically located on local or county roads that are not part of the state or national highway system, require thoughtful pavement design to meet local standards while still considering the requirements of vehicular traffic and durability.

For off-system bridges, the standard pavement design includes 8" of flex base, followed by prime and a seal coat. This design ensures durability and proper structural support for the roadway. The pavement design for off-system bridges includes several key steps, each of which must be approached with care to ensure that the pavement is suitable for its intended use. These steps generally involve assessing traffic loads, environmental conditions, and geotechnical factors to determine the most appropriate pavement structure.

7.12.1. Traffic Load Analysis

One of the first considerations in designing the pavement for an off-system bridge is evaluating the expected traffic loads. This includes:

- **Vehicle Types and Frequencies:** A detailed analysis of the types of vehicles that will use the bridge (e.g., passenger cars, trucks, buses) and their traffic volumes is essential. For off-system bridges, the design will generally focus on local traffic, but it may still need to account for larger vehicles, including agricultural machinery or delivery trucks.
- **Load Rating:** The off-system bridge must be load-rated based on the traffic that will cross it. The pavement design should account for both dynamic and static loads from vehicles, considering the bridge's capacity and the weight distribution of vehicles.

7.12.2. Geotechnical Evaluation

Pavement performance and longevity are heavily influenced by the underlying soil and subsurface conditions. A thorough geotechnical investigation is necessary to gather data on:

- **Soil Properties:** Soil classification, compaction, and shear strength characteristics are determined through testing (e.g., soil boring, Standard Penetration Test, etc.).
- **Subgrade Preparation:** Based on the soil analysis, the subgrade may need to be treated or stabilized to provide a stable foundation for the pavement. This might include modifications such as soil compaction or the use of geosynthetics to prevent shifting or settling.

- **Groundwater Considerations:** If the bridge is situated in an area with high groundwater levels, proper drainage systems must be designed to prevent water accumulation beneath the pavement, which can weaken the structure.

7.12.3. Pavement Structure Design

The pavement structure consists of several layers, each with a specific function. The design will specify the thickness and material of each layer, which typically include:

- **Subgrade:** The natural soil or fill material underneath the pavement. Its properties significantly impact the design of the upper layers.
- **Subbase Layer:** This layer, usually made from gravel or crushed stone, provides additional support and facilitates drainage. It is typically placed above the subgrade.
- **Base Layer:** This layer serves as the primary load-distributing layer and is typically made from materials such as crushed stone, gravel, or stabilized soil. The base layer helps reduce the pressure on the underlying subgrade.
- **Surface Layer:** The top layer of the pavement, typically composed of asphalt or concrete, provides the smooth driving surface and protects the underlying layers from weathering and traffic-induced wear. The material choice depends on local conditions and the desired level of durability.

Pavement thickness and material choices depend on factors such as traffic loads, local soil conditions, climate, and the expected lifespan of the bridge. For example, a heavily trafficked off-system bridge might require a thicker or reinforced pavement structure, while lighter traffic might allow for thinner layers.

7.12.4. Drainage Design

Effective drainage is crucial to the longevity of both the pavement and the bridge structure. Proper drainage ensures that water does not accumulate beneath the pavement or around the bridge foundation, which can lead to soil erosion, undermining of the pavement layers, and eventual pavement failure. Key drainage elements include:

- **Surface Drainage:** Ensuring the pavement surface is crowned or sloped to direct water toward the edges, where it can flow into ditches, culverts, or stormwater systems.
- **Subsurface Drainage:** Installing drainpipes or geotextiles to allow water to flow beneath the pavement layers without causing damage to the subgrade or base materials.

7.12.5. Environmental Considerations

Environmental conditions such as temperature extremes, freeze-thaw cycles, and local climate must be taken into account when designing the pavement for off-system bridges. In colder climates, for example, frost heave can be a significant concern, requiring additional attention to the base and subgrade layers to prevent damage. Other factors to consider include:

- **Climate Data:** Local temperature, precipitation, and freeze-thaw cycles influence the materials and thickness of the pavement.
- **Water Table:** High groundwater levels may require special drainage techniques or the use of more durable pavement materials to prevent deterioration.

7.12.6. Pavement Performance and Maintenance

Once the pavement is designed and constructed, it is essential to plan for ongoing maintenance to preserve its functionality over time. This may include regular inspections for signs of cracking, rutting, or surface wear, and prompt repairs to prevent further damage. Additionally, for off-system bridges, maintenance often includes:

- **Rehabilitation or Overlay:** As the pavement ages, it may require resurfacing with an additional layer of asphalt or other materials to restore surface smoothness and durability.
- **Crack Sealing and Patching:** Routine crack sealing helps prevent water infiltration, which can weaken the pavement structure and accelerate deterioration.

7.12.7. Design Standards and Guidelines

The design of pavement for off-system bridges should conform to local and national guidelines and standards, such as:

- **AASHTO Pavement Design Guide:** The American Association of State Highway and Transportation Officials (AASHTO) provides guidelines for pavement design based on traffic volume, load-bearing capacity, and environmental conditions.
- **Local and State DOT Guidelines:** State and local departments of transportation (DOTs) may have additional standards and requirements for pavement design, particularly for off-system bridges, which may not always align with the stricter standards used for interstate or state highways.

The pavement design for off-system bridges is a crucial process that requires a comprehensive understanding of traffic loads, geotechnical conditions, drainage needs, and environmental factors. By following standard design procedures and incorporating appropriate materials and thicknesses, engineers can ensure that the pavement will provide safe, durable, and cost-effective performance for local communities. Additionally, careful planning for maintenance and future rehabilitation can extend the lifespan of the pavement and reduce the need for costly repairs or replacements.

7.13 Utility Construction Project

In the Yoakum District, the handling of joint bid utilities is critical for ensuring smooth coordination between utility work and construction activities. One of the first steps is to ensure that joint bid utility activities are integrated into the project schedule. Designers must schedule these activities alongside the main construction work to avoid delays and conflicts. Each utility conflict, whether it involves relocations, installations, or coordination issues, should be clearly outlined in the project's contract time determination schedule. To mitigate risks, a 6-month buffer should be included between the utility clearance and the impact on construction activities, providing enough time for unforeseen delays.

The Utility Construction Management Plan (CMP) plays a vital role in coordinating utility work with the construction schedule. The design Project Manager (PM) should facilitate a smooth transition to the construction PM by arranging a meeting to discuss the utility management plan. The construction PM will assume responsibility for managing utilities during the construction phase. It is essential that utility conflicts are addressed early in the design phase, so that appropriate resolutions can be implemented to avoid delays during construction. To ensure clarity in scheduling, each utility conflict must have a specific activity, and potential utility issues should be identified in advance to avoid disruptions during construction.

By thoroughly incorporating joint bid utility work into the project schedule, traffic control plans, erosion control measures, and detailed notes for managing live utilities, designers ensure that the utility relocation and installation process goes smoothly. This approach minimizes delays and ensures that utility conflicts are managed efficiently throughout the course of the project. Proper planning and coordination between the design and construction teams are essential for the successful integration of utility work with the overall construction process.

7.14 Full Rehabilitation Project Pavement Design

The pavement design for a Full Rehabilitation Project in the Yoakum District follows a structured approach to ensure long-lasting and high-performance roadways. The project will focus on rehabilitating the existing pavement structure using a combination of new and existing materials, including lime-treated base (CTB), surface treatments, and asphalt layers. The quantities and specifications of these materials will be calculated to match the roadway's surface area, with particular attention to subgrade preparation and widening requirements.

In summary, the full rehabilitation of the roadway involves a comprehensive design approach that integrates new lime-treated base, surface treatments, and asphalt layers. The use of both existing and new base material ensures the pavement structure is strengthened without unnecessary over-excavation. The final surface treatment provides a smooth and durable finish capable of withstanding high traffic volumes, while the widening of the subgrade and the careful cutting of the pavement edges ensure a proper foundation for the new layers.

The project also requires attention to the specific quantities of materials, with lime content and Flex Base being carefully calculated to meet the structural requirements. Special attention is given to the edge conditions, ensuring that the new pavement transitions smoothly into the existing roadway, providing a durable, long-lasting rehabilitation for the entire project area. All work should be coordinated closely with the project specifications to meet both design and construction standards for a successful and sustainable outcome.

7.15 Cable Barrier Safety Project

For Cable Barrier Safety Projects, refer to Item 543 in the guide for detailed specifications and requirements related to cable barrier installation and maintenance. This item outlines the construction and installation practices, as well as the safety standards necessary to ensure the proper function of cable barriers in highway settings. Cable barriers are a key safety feature for reducing fatalities and injuries by redirecting errant vehicles back onto the roadway.

7.16 Traffic Signals

Certain cities within the TxDOT Yoakum District require city review and approval for traffic signal installations. Specifically, any signal work within the city limits of Victoria, Wharton,

and Sealy must be reviewed and coordinated with the respective city's traffic management department. It's essential to follow the local city's guidelines and ensure that the traffic signal design adheres to both TxDOT and municipal requirements to ensure smooth integration into the existing traffic control systems.

7.17 Aesthetics for Bridge and Wall Designs

When designing the aesthetics for bridges and retaining walls within the Yoakum District, it is essential to coordinate with the Area Engineer to verify the appropriate theme for the specific region or project. The Area Engineer will help ensure that the selected design reflects the desired visual and cultural identity of the area while maintaining functionality and safety standards.

These themes are inspired by the natural landscape of Texas and include elements that reflect the region's rolling hills, scenic vistas, and historical landmarks.

In all cases, these aesthetic themes should be verified and confirmed with the respective Area Engineer before finalizing the design to ensure alignment with local standards and community expectations. Properly incorporating these regional themes enhances the visual appeal of transportation infrastructure.

7.18 Safety Rest Area Project

This design guide provides specifications and standards for the planning, design, and construction of rest areas within the Texas Department of Transportation (TxDOT) Yoakum District. The guide is structured to ensure that rest areas are safe, accessible, sustainable, and adhere to both the district and state standards.

7.18.1. General Guidelines

7.18.1.1 Purpose and Need

Rest areas provide safe, clean, and accessible places for drivers and passengers to stop, rest, and refresh. They are essential for reducing fatigue-related accidents and supporting travelers' well-being. These facilities must balance functionality with aesthetics and sustainability.

7.18.2 Location Criteria

7.18.2.1 Site Selection

- **Spacing:** Rest areas should be located approximately every 50–75 miles along interstate highways or major freeways, based on traffic volume and geography.
- **Access Roads:** Access to rest areas must be from main highways or interstates via clearly marked ramps and exit lanes. The entrance should allow for easy deceleration and the exit should allow for smooth acceleration back onto the highway.
- **Topography:** Avoid floodplains, wetlands, or areas of steep terrain. The site should allow for manageable grading and minimal environmental disruption.

7.18.2.2 Environmental Sensitivity

- **Floodplain Considerations:** Rest areas should not be located within the 100-year floodplain unless no other alternative exists, and floodplain management practices are followed.
- **Soil Type and Stability:** Site evaluation must ensure that the soils can support necessary infrastructure (e.g., parking lots, roads, utilities) with appropriate compaction and stabilization.

7.18.3. Roadway and Traffic Considerations

7.18.3.1 Access Road Design

- **Design Speed:** The access road must be designed for speeds of 25–30 mph depending on site conditions, traffic volume, and expected use.
- **Width of Access Roads:** The roadway for vehicles entering and exiting the rest area should have a minimum width of 12 feet per lane.
- **Curb Radius:** The minimum curb radius for vehicle turning movements should be 50 feet to accommodate large vehicles like trucks and RVs.
- **Pavement Design:**
 - ✓ **Hot Mix Asphalt (HMA):** For all access roads and parking areas. The typical thickness should be 6 inches with a base of 12 inches of crushed stone.
 - ✓ **Portland Cement Concrete (PCC):** If required, use concrete with a compressive strength of 4,000 psi for high-use areas like truck parking spaces or heavy loading zones.

7.18.3.2 Parking Design

- **Truck Parking:**
 - Truck and RV spaces should have a minimum size of 12 feet by 65 feet.

- Parking lots must be designed with adequate turning radii to accommodate large vehicles. Minimum radii for truck access lanes should be 50 feet.
- Passenger Vehicle Parking:
 - Standard passenger vehicle spaces should be 9 feet by 18 feet.
 - Provide additional accessible parking spaces in compliance with ADA guidelines.
- Pedestrian Access:
 - Provide clear, ADA-compliant pedestrian pathways with a minimum width of 5 feet connecting parking areas to amenities such as restrooms, picnic areas, and vending machines.

17.18.4. Stormwater Management and Sustainability

17.18.4.1 Stormwater Control

- Drainage Design:
 - ✓ All parking areas, restrooms, and other impermeable surfaces must be graded to ensure proper drainage.
 - ✓ Use permeable pavement in areas where possible to reduce runoff and support infiltration. Permeable pavers, asphalt, or concrete can be used in parking areas with low traffic volume.
 - ✓ Install stormwater catch basins with sediment traps to filter runoff water.

17.18.4.2 Green Infrastructure

- Landscaping:
 - ✓ Native Plants: Use native, drought-resistant species for landscaping, including trees, shrubs, and ground cover. These plants should be chosen based on local environmental conditions and should reduce the need for irrigation.
 - ✓ Irrigation Systems: If irrigation is needed, use low-flow irrigation systems or drip irrigation to minimize water usage.
- Solar Power:
 - ✓ Consider the installation of solar panels for powering lighting in parking areas and restrooms.

17.18.5. Signage and Traffic Control

17.18.5.1 Signage

- Directional Signs:
 - ✓ Provide clear, illuminated signs directing drivers to the entrance and exit of the rest area. Ensure signs are visible from a distance, with a font size large enough for visibility at high speeds.
 - ✓ Restroom Signs: Clearly mark restroom locations using universally understood symbols. Include tactile and braille signage where applicable.

17.18.5.2 Traffic Control Devices

- Speed Limits:
 - ✓ Establish speed limits within the rest area to control traffic flow, with recommended speeds not exceeding 25 mph for access roads.
 - ✓ Provide advance warning signs of speed limits on approach roads.
- Lighting:
 - ✓ Provide high-efficiency LED lights along access roads, parking lots, walkways, and around restrooms to ensure safety during nighttime use.
 - ✓ Ensure lighting levels meet minimum 2-foot candles for pedestrian areas and 5-foot candles for vehicle areas.

17.18.6. Maintenance and Operational Considerations

17.18.6.1 Maintenance Schedule

- Routine Inspections: Rest areas should be inspected regularly (minimum quarterly) for wear and tear, and all facilities should be maintained and cleaned on a daily basis.
- Restroom Cleaning:
 - ✓ Restrooms should be cleaned and disinfected 3-4 times per day depending on traffic volume.
 - ✓ Inspect and maintain plumbing to prevent clogs or leakage.
- Landscaping:
 - ✓ Mowing, trimming, and maintenance of landscaping should be performed bi-weekly during the growing season.

17.18.6.2 Emergency Services

- Emergency Call Boxes: Install emergency call boxes with direct connection to local emergency services in high-use rest areas or those in remote locations.
- First Aid Kits: Rest areas should be equipped with first aid kits available in accessible locations, such as near restroom entrances.

17.18.7. Accessibility and Safety

17.18.7.1 ADA Compliance

- Accessible Parking: Ensure that 5% of parking spaces are designated as ADA-compliant.
- Accessible Pathways: Ensure all walkways to restrooms, vending areas, and picnic spaces are accessible to individuals with mobility impairments. Walkways should have a minimum width of 5 feet and a slope of no more than 1:12.

17.18.7.2 Safety Features

- **Lighting:** Install motion-sensor lighting in isolated areas such as restrooms and parking lots to improve security while minimizing energy consumption.
- **Surveillance:** Consider integrating CCTV cameras at strategic locations, especially in high-use rest areas, to ensure safety and monitor for potential vandalism or criminal activity.

17.18.8. Future Considerations

17.18.8.1 Expansion

- **Flexibility:** When designing the rest area, consider future expansions for additional amenities, such as more parking spaces, seating areas, or additional restroom facilities.
- **Infrastructure Upgrade:** Design all utilities (water, sewer, electrical) with future capacity in mind.

17.18.8.2 Sustainability Initiatives

- **Electric Vehicle (EV) Charging Stations:** Consider incorporating EV charging stations into the rest area design to accommodate the growing demand for electric vehicles.

By adhering to these design principles and specifications, TxDOT Yoakum District ensures that rest areas are not only safe and functional but also sustainable, accessible, and comfortable for all users. This guide should be followed closely to meet the evolving needs of travelers while maintaining high standards of environmental stewardship and operational efficiency.

Review and Approval

8.1 Item-by-Item Guide

The Item-by-Item Guide is an essential resource for ensuring that all design elements are accurately referenced and applied according to district-specific requirements. This guide provides detailed information on each item, including specifications, guidelines, and requirements for construction and material use.

District-Specific General Note: The Yoakum District has established specific guidelines essential for the successful execution of construction projects. These district-specific notes offer valuable context, clarifications, and, in some cases, additional requirements for certain project components. It is crucial to reference these notes when preparing plans, as they provide critical insights into the district's preferences and regulatory requirements.

Contact List: For any inquiries or requests for further information pertaining to specific items, a contact list is provided to facilitate communication with the relevant personnel. This list includes district engineers, project managers, and subject matter experts who can offer clarifications or provide assistance with particular project requirements. It is essential to consult the appropriate contacts for matters concerning items, specifications, or design clarifications to ensure that the project remains in compliance with established guidelines and progresses in accordance with local regulations.

For detailed references and to avoid confusion, always consult the contact list and the district-specific general notes when uncertain about any part of the project design or implementation process. This ensures smooth communication and adherence to all relevant standards.

Glossary

Glossary

A+B Bidding: A method of bidding in which the construction cost (A) and time (B) are combined, encouraging contractors to complete the project more quickly.

Accelerated Pavement Testing: process used to simulate the long-term effects of traffic and weather on roadways by applying controlled, accelerated traffic loads to pavement surfaces. This testing helps engineers evaluate the durability, performance, and longevity of different pavement materials and designs.

ADT (Average Daily Traffic): The average number of vehicles passing a specific point on a roadway per day, used to help assess traffic volume and plan capacity.

Alignment: The horizontal and vertical layout of a roadway, including its direction, curvature, and grade.

As-Built: Final construction plans that reflect the actual conditions of the completed project, including any modifications made during construction.

Asphalt Binder: The asphalt component used in hot mix asphalt (HMA), which acts as the adhesive that binds the aggregate particles together. The binder is the most critical component in determining the strength and durability of the pavement.

Asphalt Concrete (AC): A type of pavement material made from a mixture of asphalt binder and aggregates used for road surfaces.

Ballasted Track: A type of railroad track system in which the rails are supported by a foundation of crushed stone or gravel. The ballast helps distribute the weight of passing trains, provides drainage, and holds the track in place.

Base Course: The layer of material (usually crushed stone or gravel) placed beneath the pavement structure to provide strength, support, and drainage. It serves as a foundation for the surface layers of the roadway and contributes to its overall stability.

Bridge Deck: The top surface of a bridge that carries traffic loads. The deck is the horizontal structure of the bridge that directly supports vehicle and pedestrian traffic.

Base Course: The layer of material (usually crushed stone or gravel) placed beneath the pavement structure to provide strength, support, and drainage. It serves as a foundation for the surface layers of the roadway and contributes to its overall stability.

Base Flood Elevation (BFE): The elevation at which floodwaters are expected to rise during a 100-year flood event.

Bridge Hydraulics: The study of water flow and drainage around and under a bridge to prevent flooding or erosion of the foundation.

Bridge Load Rating: The calculation of the weight-bearing capacity of a bridge, used to determine whether it can safely accommodate specific traffic loads.

Bridge Rail: Safety barriers placed on the edges of bridges to prevent vehicles from leaving the roadway.

Buffer Zone: A designated area, typically landscaped, that separates different land uses or transportation features. Buffer zones help reduce noise, improve safety, and provide visual appeal along roadways or around developments.

Clear Zone: The area beside the roadway, free of obstacles, that allows vehicles to recover safely if they leave the roadway.

Coffer Dam: It is a temporary structure built to divert or control the flow of water, creating a dry work environment for construction projects, typically in rivers, lakes, or other bodies of water. It allows work such as excavation, foundation placement, or repair to be done below water level.

Contract Time Determination Schedule: A detailed timeline for completing various phases of a construction project, used to track progress and ensure timely completion.

Crossover: A temporary roadway structure or a shift in traffic patterns used to reroute traffic around work zones or construction areas. Crossovers are often used in highway construction to maintain traffic flow during repairs or upgrades.

Cross Section: A diagram showing a slice or cut through the roadway at a specific location, used to illustrate the road's structure and dimensions.

Culvert: A structure, typically a pipe, placed under a roadway to allow the flow of water from one side of the road to the other. Culverts are used to manage stormwater runoff and prevent flooding by providing drainage across roads, highways, and railways.

Culvert Design: The design of underground drainage structures, typically pipes or tunnels, to channel water under a roadway.

Curb Ramp Design: Designs for ramps that provide accessible pathways for pedestrians with disabilities, typically located at intersections.

Decel/Accel Lanes: Decel/Accel Lanes are lanes designed for vehicles to slow down (deceleration lane) or speed up (acceleration lane) when entering or exiting a highway.

Deceleration Lane: A lane designed to allow vehicles to slow down safely when exiting a highway or road.

Detour Route: An alternative route established during road construction or maintenance work to redirect traffic around a closed or obstructed roadway. Detour routes are designed to minimize disruption and maintain traffic flow during periods of construction or emergency closure.

Design Speed: The speed at which a road is designed to be traveled under ideal conditions, influencing road geometry, curvature, and other features.

Dynamic Message Signs (DMS): Electronic traffic signs that display real-time information to drivers, such as traffic conditions, detour information, or safety alerts. DMS are typically part of Intelligent Transportation Systems (ITS) and are used to improve traffic management.

Embankment: A raised structure made of earth, rock, or other materials that supports a roadway or railway. Embankments are used to elevate roads above the surrounding terrain, protect against flooding, or create stable foundations for infrastructure.

Environmental Impact Statement (EIS): A report assessing the potential environmental effects of a project, required for projects that significantly affect the environment.

Environmental Layouts: Design plans that show environmentally sensitive areas, such as wetlands, floodplains, and wildlife habitats, and how they are considered in the project design.

Estimate: An approximation of the costs, materials, and resources required for the completion of a construction project.

Field Visit and/or Surveying: On-site activities to collect data, assess existing conditions, and verify project requirements through direct observation or measurements.

Flexible Pavement: A type of pavement structure that is designed to distribute traffic loads to the underlying layers through bending. Flexible pavements are typically constructed using layers of bituminous materials (such as asphalt) and are more adaptable to changes in load distribution and temperature than rigid pavements.

Floodplain Considerations: Factors to be addressed when designing projects located in floodplain areas, ensuring compliance with regulations and minimizing flood risk.

Gabion: Gabions are wire mesh baskets or containers filled with rocks or other materials, commonly used for erosion control, slope stabilization, and landscaping, as well as for creating retaining walls or flood barriers.

Guardrail: A safety barrier placed along roadways to prevent vehicles from leaving the road or colliding with obstacles, particularly in high-risk areas like curves or bridges.

Highway Capacity Manual (HCM): A publication providing methodologies for assessing the capacity and performance of roadways based on traffic volume, road design, and other factors.

Hot Mix Asphalt (HMA): A type of asphalt mixture that consists of asphalt binder and aggregates (such as sand, gravel, or crushed stone) mixed together at high temperatures (typically between 300°F and 350°F). HMA is commonly used for surface treatments, overlays, and new pavement construction due to its durability, strength, and ability to withstand high traffic loads.

Hydraulics: The study of the flow and control of water, especially in drainage systems, stormwater management, and bridge design.

Hydrology: The science of water movement and distribution, which is essential in transportation planning, particularly for drainage and flood management.

Intelligent Transportation Systems (ITS): Systems that use advanced technology to manage traffic flow, reduce congestion, and improve safety through real-time data and communication.

Interchange: A grade-separated intersection of two or more roads or highways, allowing vehicles to change roads without crossing traffic paths. Interchanges are commonly used on highways to reduce congestion and improve traffic flow.

Intersection Control Evaluation (ICE): A systematic analysis of the best control measures for intersections, including the use of signals, signs, and lane configurations.

Item-by-Item Guide: A reference that provides details and specifications for each bid item included in the construction contract, assisting in project planning and execution.

Joint-Bid: A joint bid is a proposal submitted by two or more companies or contractors who collaborate to bid on a project, sharing responsibilities, resources, and risks to increase their chances of winning the contract.

Kirby-Kirpich Method: It is a hydrological formula used to estimate the time of concentration (T_c) for a watershed, which is the time it takes for runoff to travel from the furthest point in the watershed to the outlet, based on the length of the flow path and the slope of the terrain. This method is commonly used in stormwater management to design drainage systems for small to medium-sized watersheds.

Lateral Support: The resistance provided by the soil or foundation material that prevents a structure, such as a retaining wall, embankment, or roadway, from sliding or moving sideways. Lateral support is crucial for the stability of embankments and other earthwork structures.

Limited Access Highway: A type of highway that restricts access to certain entry and exit points, typically through ramps or interchanges. Limited access highways are designed to facilitate the rapid and safe movement of traffic, especially in urban and suburban areas.

Mill and Inlay: A process for resurfacing roads, where the top layer of pavement is milled (removed) and replaced with a new layer (inlaid) to restore the surface condition.

Mill and Overlay: A pavement maintenance technique that involves removing (milling) a portion of the existing pavement surface and then applying a new overlay to restore or improve the road's condition. This method is commonly used to address surface distresses and extend the life of the pavement.

Mobility Project: Projects designed to improve transportation efficiency, often by increasing capacity, reducing congestion, or enhancing flow.

Non-Overlay Project: Projects where road resurfacing (overlay) is not performed, but other improvements such as widening or structural repairs are made.

Notch: A notch is a small, V-shaped or U-shaped cut or indentation made in a surface or material, often used for alignment, fitting, or structural purposes, such as in road construction to provide space for transitions or drainage.

Overlay: A method of resurfacing a road by adding a new layer of asphalt or other material over the existing surface.

Pavement Design: The process of designing the composition and structure of a roadway surface to accommodate the anticipated traffic loads and environmental conditions.

Pavement Markings: Lines, symbols, and words painted or applied to the surface of a roadway to convey information to drivers and regulate traffic. Pavement markings include lane lines, crosswalks, and symbols indicating turn lanes, speed limits, and pedestrian crossings.

Pavement Transition: The gradual change in road structure from one type of pavement or condition to another, such as from old pavement to a newly overlaid surface.

Pedestrian Infrastructure: Design elements aimed at supporting pedestrian movement and safety, including sidewalks, crosswalks, ramps, and pedestrian signals.

Pedestrian Safety: The implementation of measures such as proper crosswalks, pedestrian signals, and traffic calming features to ensure the safety of people walking along roadways.

Prime Coat: A layer of bituminous material (such as asphalt or oil) applied to a prepared surface before applying a subsequent layer, such as an asphaltic binder or wearing course. The prime coat serves to bond the new layer to the surface and helps prevent moisture from entering the subgrade, which can compromise the pavement structure.

Rehabilitation: The process of restoring or improving an existing structure, such as a bridge or roadway, to extend its service life and improve its safety and functionality. Rehabilitation may involve resurfacing, structural repairs, or other upgrades.

Right of Way (ROW): A legal right to use or occupy land for transportation purposes, including roadways, utilities, and easements.

Right-of-Way Acquisition: The legal process of obtaining property or land use rights from landowners to construct transportation infrastructure such as roads, bridges, and utilities. ROW acquisition is an important part of project planning and can involve negotiation, compensation, and legal documentation.

Rigid Pavement: A type of pavement that uses concrete or other materials with high flexural strength to distribute traffic loads. Rigid pavements are typically thicker than flexible pavements and are designed to transfer loads through bending resistance.

Riprap: A layer of large stones or concrete used to protect soil from erosion.

Road User Cost (RUC): A calculation of the costs incurred by the public due to delays, detours, or other disruptions caused by construction or maintenance projects.

RUC (Runoff Coefficient) Calculation: This method is used in hydrology to estimate the amount of rainfall that will turn into runoff for a given area. It is a dimensionless factor that reflects the characteristics of the land surface, such as land use, soil type, and vegetation cover, which influence the amount of runoff generated during a rainfall event.

Safety Edge: A feature incorporated into the design of roadways, where the pavement edge is sloped or beveled to provide a safer transition from the roadway to the shoulder or ditch. This design helps reduce the likelihood of vehicles running off the road and improves safety.

Safety Score Tool: A tool used to assess the safety of a proposed design or project by considering factors like traffic patterns, road geometry, and accident history.

Seal Coat: A protective treatment applied to an existing roadway surface to seal cracks, prevent water infiltration, and restore the surface's integrity. Seal coats consist of a layer of asphalt binder sprayed over the surface, followed by the application of fine aggregate. It helps extend the lifespan of the road and improve skid resistance.

Setback: The required distance between a proposed construction or structure (such as a road, building, or utility) and the property line, existing infrastructure, or other designated areas. Setbacks ensure that construction does not interfere with neighboring properties or environmental concerns.

Side Slope: The inclined surface or slope adjacent to a roadway or embankment, often designed for drainage and stability. Side slopes are critical for ensuring proper water flow away from the roadway and reducing erosion. They are typically expressed as a ratio of horizontal distance to vertical height (e.g., 2:1 side slope, meaning for every 2 feet of horizontal distance, the slope rises or falls by 1 foot).

Shoulder: The portion of the roadway adjacent to the traveled way, used for emergency stops and vehicle breakdowns.

Smart Work Zone: A work zone that uses advanced technology such as sensors, cameras, and variable message signs to improve traffic flow and safety during construction.

Stormwater Pollution Prevention Plan (SWP3): A plan required for construction projects to manage stormwater runoff and prevent pollution during construction activities.

Striping: The process of painting road markings, including lane lines, crosswalks, and other traffic control markings, to guide and regulate traffic.

Subgrade: The soil prepared to support a pavement structure

Superstructure: The part of a bridge that carries the traffic load and spans the obstacle being crossed.

Superelevation: Superelevation refers to the banking or tilting of a roadway curve, where the outer edge is raised higher than the inner edge, to counteract centrifugal force and help vehicles navigate the curve safely at higher speeds.

Surface Treatment: The application of materials such as seal coat or slurry seal to a pavement surface to restore its condition and extend its service life.

Temporary Traffic Control Plan (TCP): A plan that outlines how traffic will be rerouted and managed during construction, including detours, road closures, and signage.

Traffic Analysis: The process of evaluating traffic conditions, including flow, congestion, and safety, to inform design decisions and improve transportation systems.

Traffic Control Devices (TCDs): Signs, signals, barriers, and other devices used to direct, regulate, or warn drivers on the road.

Traffic Control Plan (TCP): A comprehensive plan that details how traffic will be managed during a project, including lane closures, detours, and temporary signage.

Traffic Data: Data collected from roadways and intersections to assess traffic volume, speed, and patterns, which is essential for traffic modeling and planning.

Traffic Impact Analysis (TIA): A study conducted to assess the effects of a proposed development or construction project on the surrounding transportation network. A TIA evaluates changes in traffic volumes, congestion, safety, and access and recommends measures to mitigate negative impacts.

Traffic Signal Design: The process of designing and installing traffic signals at intersections to control the flow of vehicles and pedestrians.

Transverse Slope: The slope or crossfall of a road, designed to direct water off the pavement surface and into drainage systems or the shoulder. The transverse slope is typically 2% to 4% for roads, ensuring proper drainage and preventing water accumulation on the road surface.

Typical Section: A cross-sectional view of a roadway, representing a typical or standard design for a given segment of the road. It shows the arrangement of lanes, shoulders, ditches, slopes, and other features that make up the roadway's structure. Typically used to define the general layout of a roadway for design and construction purposes.

Utility Relocation: The process of moving utilities (e.g., water, sewer, electrical lines) to make room for roadway improvements, construction, or expansion.

Utility Corridor: A designated area for the installation of utilities such as water, sewer, gas, or electrical lines. Utility corridors are planned to avoid interference with transportation infrastructure and ensure safe and efficient utility services.

Vertical Profile: A representation of the changes in elevation along the length of a road, important for determining road grades and drainage requirements.

Water Quality Management (WQM): The management and design practices aimed at protecting water resources from contamination during construction projects. WQM includes strategies for controlling runoff, preventing sedimentation, and mitigating other environmental impacts during and after construction.

Widening Project: A construction project that expands an existing road to accommodate more lanes, improve safety, or enhance traffic flow.